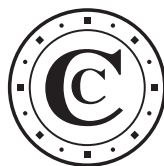


Cour des comptes



ENTITIES AND PUBLIC POLICIES

THE NATIONAL STRATEGY FOR ARTIFICIAL INTELLIGENCE

Public policy on AI:
consolidating its successes,
broadening its scope

Thematic public report

November 2025

Summary

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Procedures and methods

The reports of the Court of Auditors are produced by one of the six thematic chambers¹ that make up the Court, or by a panel comprising several chambers and/or several regional or territorial chambers of accounts.

Three fundamental principles govern the organization and activities of the Court and the regional and territorial chambers of accounts, including the performance of their audits and investigations and the preparation of public reports: independence, adversarial proceedings and collegiality.

The institutional **independence** of the Financial Jurisdictions and the statutory independence of their members guarantee that the audits carried out and the conclusions drawn are based on complete freedom of assessment.

Adversarial proceedings means that all findings and assessments made during an audit or investigation, as well as all observations and recommendations made thereafter, are systematically submitted to the heads of the administrations or bodies concerned; they can only be made final after the responses received have been taken into account and, where appropriate, after the heads concerned have been heard.

Collegiality is involved in concluding the main stages of the audit and publication procedures. Each audit or investigation is entrusted to one or more rapporteurs. The audit report, as well as subsequent draft observations and recommendations, both provisional and final, are examined and deliberated on a collegial basis by a panel comprising at least three magistrates. One of the magistrates acts as counter-rapporteur and ensures the quality of the audits.

Except for reports produced at the request of Parliament or the Government, the publication of a report is necessarily preceded by the communication of the draft text that the Court intends to publish to the ministers and heads of the bodies concerned, as well as to other legal or natural persons directly concerned. Their responses are presented in an appendix to the report published by the Court.

**

This report is the result of an investigation conducted on the basis of Articles L. 111-2, L. 111-3 and L. 111-4 of the Code of Financial Jurisdictions, which allow the Court to conduct thematic investigations. It follows on from the public policy evaluation report published by the Court of Auditors in April 2023 on the research and higher education component of the first phase of the national strategy for artificial intelligence.

¹ The Court also has a litigation chamber, whose judgments are made public.

The opening of the investigation was notified on 21 February 2025 to the representatives of the administrations and bodies concerned, and the investigation took place from March to May 2025. It involved on-site and documentary investigations in some 20 entities, as well as in-depth interviews with around 50 people. Following the written exchange on its detailed provisional observations, the Court (Third Chamber) also held several hearings in September 2025.

**

The draft report was prepared and then deliberated on 25 September 2025 by the Third Chamber, presided over by Mr Meddah, Chamber President, and composed of Mr Vallernaud, Senior Advisor, Section President, Ms de Mazières, Senior Advisor, Section President, Ms Fontaine, Messrs Feltesse, Malcor and Aubert, Senior Advisers, Ms Haguenaer, Senior Adviser on special duty, as well as, as rapporteur, Mr Kruger, Senior Adviser, and, as co-rapporteur, Mr Mousson, Senior Adviser.

It was examined and approved on 6 October 2025 by the Public Report and Programmes Committee of the Court of Auditors, composed of Mr Moscovici, First President, Mr Hayez, General Rapporteur of the Committee, Ms Camby, Mr Bertucci, Mr Meddah, Ms Mercereau, Mr Cazé, Presidents of Chambers of the Court, Mr Fulachier, Section President, representing Mr Lejeune, President of Chamber of the Court, Mr Oseredczuk, Section President, representing Ms Thibault, President of Chamber of the Court, Mr Albertini, Ms Mouysset, Ms Daussin-Charpantier and Ms Daam, presidents of regional chambers of accounts, and Ms Hamayon, Attorney General, whose opinions were heard.

**

The public reports of the Court of Auditors are available online on the website of the Court and the regional and territorial chambers of accounts: www.ccomptes.fr.

Summary

Following a trend that began in several countries in the mid-2010s, France decided to launch a strategic review of the challenges associated with artificial intelligence (AI) and to develop a specific public policy in this area. The first phase of the National Strategy for Artificial Intelligence (*Stratégie nationale pour l'intelligence artificielle*, SNIA), conducted between 2018 and 2022, focused primarily on strengthening research in this field. A second phase, known as the acceleration phase, was announced at the end of 2021 and has been implemented mainly from 2023 onwards, with the central objective of disseminating AI throughout the economy. In February 2025, on the occasion of the Paris Summit for Action on Artificial Intelligence, the President of the Republic announced a third phase of the national strategy, the details of which were clarified in the months that followed.

The 2018-2022 phase of the strategy: strengthening AI research

The implementation of the first phase of the national strategy for artificial intelligence made it possible to initiate a public policy on AI in France, even if it was only able to cover part of the issues identified in March 2018. The management and implementation of this first phase was based on a complex interplay between numerous actors. The diverse resources allocated by the government ultimately amounted to €1.3 billion, and their monitoring proved to be deficient.

Despite several limitations, the main contribution of this phase was to help initiate the development and structuring of research and innovation in the field of artificial intelligence, with the creation of centers of excellence, the opening of essential computing infrastructures and support for the growth of AI startups in a variety of fields.

In the other areas covered by the SNIA – defense and security, the transformation of public action and the dissemination of artificial intelligence in the economy – progress has been less clear-cut. Several of the announced priorities, particularly on the key issues of training and supporting change in the economic sectors most affected by AI, have not been implemented or only to a very limited extent, at the risk of causing France to fall behind.

The 2023-2025 phase of the strategy: aiming for the diffusion of AI in the economy

Launched without prior evaluation, the second phase of the SNIA was supposed to take up the challenge of massifying and supporting the diffusion of artificial intelligence in all areas. Announced in November 2021, this phase has seen its priorities, budget and timetable change significantly due to growing constraints on public finances and the need to reallocate resources to support the development of generative AI, an issue that had not been anticipated on the eve of the “ChatGPT revolution”.

In total, the government allocated €1.1 billion over the period 2023-2025, which is one-third less than initially announced, and the slow start-up of most of the measures has resulted in a low level of budget consumption (35% as of 30 June 2025). The governance of public policy on artificial intelligence has also remained complex, despite some positive developments and several examples of successful coordination with other so-called “acceleration” strategies of the France 2030 programme.

The initial results of this second phase are beginning to emerge in several areas. Although it is still too early to fully assess the effects, the initiatives taken to strengthen the structure and excellence of research and higher education in the field of AI are producing initial results, and France’s position in this area is improving. Our country has risen from thirteenth place in the Global AI Index published in September 2024 to fifth place in September 2025. In terms of research and training in artificial intelligence, France ranks third in the world. More than 4,000 French researchers are currently working on AI.

The mobilization that this second phase has enabled in the field of generative AI has also borne fruit. At the beginning of 2023, France had only one player positioned in this type of system. In just a few months, French industry has made progress in terms of competitiveness and attractiveness, with the emergence of a dozen players operating in a wide variety of fields. The number of French AI startups has doubled since 2021: more than 1,000 of them are active in this field in 2025 and they raised nearly €2 billion in funding in 2024. Sixteen French startups valued at over \$1 billion (unicorns) incorporate artificial intelligence into their value proposition, and several major French groups are increasing their offering and investment in AI research. France is the leading European country in terms of the number of foreign investment projects in artificial intelligence, and in terms of hosting research and decision-making centers for Big Tech AI companies.

Efforts to develop computing infrastructure have continued, with the expansion of installed capacity and investment in a new-generation supercomputer. Significant progress has also been made on the issues of frugality and trust. Finally, France is no stranger to the acceleration of European policy on AI and, more broadly, to the fact that international forums are addressing key issues of governance and regulation of the development of artificial intelligence. The success of the AI Action Summit held in Paris in February 2025 confirmed France’s special place on the international stage.

However, alongside these successes, several equally important areas have been neglected. The challenge of massifying and supporting the dissemination of artificial intelligence beyond the circle of specialists – businesses, public administrations, students, citizens – has so far received too little attention, even though it was at the heart of the ambitions set out for this phase of the SNIA and the years 2023-2025 were critical in this regard. As a result, the priority of supporting business demand for artificial intelligence solutions has only benefited from very modest measures, and the expected acceleration and massification of the diffusion of artificial intelligence in the economy has not taken place. The delay in adapting all initial and continuing training programmes to AI has not been made up either, even though this is an area where the stakes are considerable and the risks high. Essential projects concerning schools and universities have yet to be designed and implemented. The transformation of public action through artificial intelligence, which has also not been a priority, has been very disappointing too: despite isolated initiatives, the administration is generally lagging behind. Finally, actions targeting regions and the general public have not been a priority for the SNIA to date, even

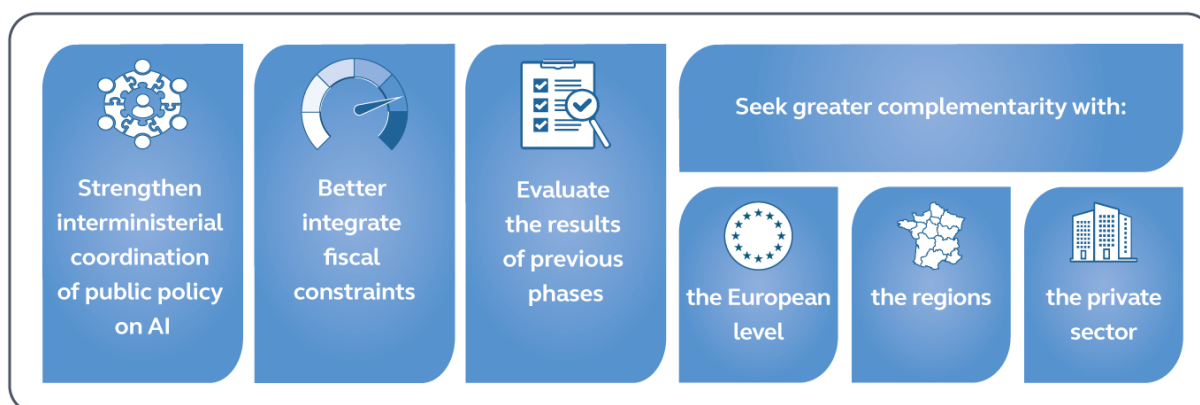
though they appear all the more necessary as the impacts of this general-purpose technology accelerate, intensify and become more widespread.

The outlook: consolidate the successes of public policy on AI and broaden its scope

A third phase of the SNIA was launched in February 2025, with the aim of accelerating the diffusion of artificial intelligence in businesses.

Artificial intelligence is no longer the same issue it was in 2018, when the SNIA was launched. The revolution brought about by this general-purpose technology is reaching a magnitude that few other technological breakthroughs in history can match. AI is no longer a matter for specialists alone; it affects all fields of knowledge, the economy, and society, and is becoming an essential priority for public policy. Successfully achieving the scale change required by the artificial intelligence revolution requires several prerequisites to be met:

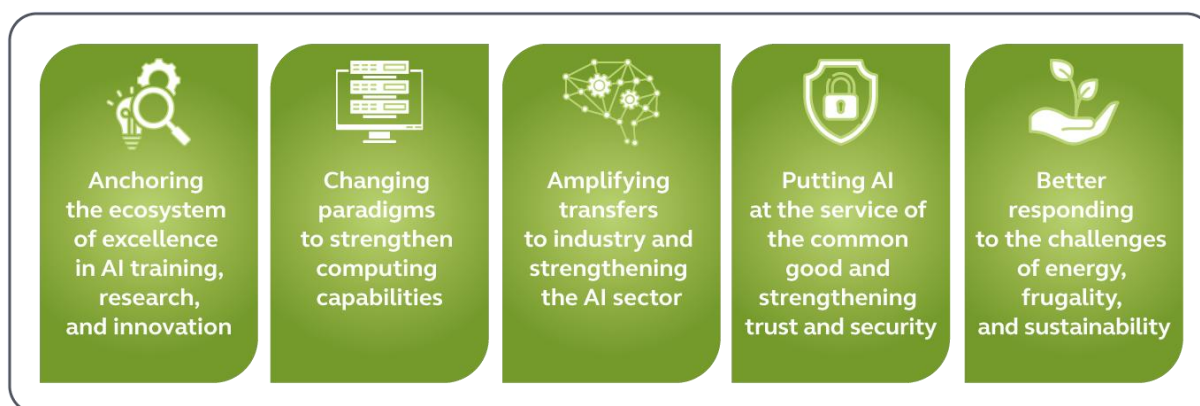
Prerequisites for successfully scaling up AI



Source: Court of Auditors

Given the rapid pace of change in the AI landscape, it would be a mistake to assume that the priorities on which SNIA has achieved initial success no longer require attention. Public policy on artificial intelligence must seek to expand its areas of excellence and aim for even more structural transformations and impacts. Five key areas, which have been under development since 2018, should therefore be further explored:

Areas for further development for more structural transformations



Source: Court of Auditors

Finally, the first two phases of the SNIA have blind spots in several key areas, or at least have failed to achieve results commensurate with the challenges. This is particularly the case in projects that require the involvement of a wide range of stakeholders and links with other public policies. Five critical challenges, which have not been sufficiently addressed to date, must be placed at the heart of public policy on AI in the coming years:

Critical challenges to be placed at the heart of public policy on AI



Source: Court of Auditors

The national strategy for artificial intelligence has created real momentum since its launch in 2018. France has managed to climb to the top of the European pack, including in the latest and most widely used generative AI technologies. It can compete with its rivals, the United States and China set apart, in most innovations related to artificial intelligence, and its visibility on the international stage, as confirmed by the Paris summit in February 2025, is real.

The Court of Auditors has made ten recommendations, the implementation of which would enable public policy on artificial intelligence to scale up, build on the initial successes of the SNIA, and overcome the limitations and shortcomings identified.

It is on this condition that France, in close cooperation with the European Union and local authorities, drawing on the training-research-innovation ecosystem as well as businesses and investors, even if it means changing its operating methods, will continue on its path to excellence and succeed in embracing all the dimensions that the AI revolution is set to affect, in the service of the common good and with a view to guaranteeing national sovereignty.

Recap of recommendations

The Court makes the following recommendations:

1. By the end of 2025, strengthen interministerial oversight of public policy on AI by establishing an *ad hoc* general secretariat, and better reconcile the necessary ambition of this policy with the challenges of efficiency and effectiveness by conducting an in-depth assessment of results and seeking greater complementarity with the European level, regions and the private sector (*Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, Ministry of Higher Education and Research, National Institute for Research in Computer Science and Automation*).

Adopt, by the next AI summit in February 2026, a strategy with a view to:

2. Ensure long-term excellence in training, research and innovation in the field of AI by better defining needs, giving stakeholders greater responsibility, reaffirming the purpose of public research and promoting mobility with the private sector (*Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, Ministry of Higher Education and Research, National Institute for Research in Computer Science and Automation*).
3. Increase, in coordination with the European Union, the computing capacities for AI and redefine, through new forms of partnership between the public and private sectors, the conditions for financing, building and operating these infrastructures, which guarantee open access to all users (*Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, Ministry of Higher Education and Research, National Institute for Research in Computer Science and Automation, National Centre for Scientific Research*).
4. Strengthen the transfer of research to industrial development and support the growth of AI companies, including by leveraging public procurement and strengthening the monitoring of industrial acquisitions in the AI sector (*Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, Ministry of Higher Education and Research, National Institute for Research in Computer Science and Automation*).
5. Implement and pursue commitments to trustworthy, frugal and sustainable AI, including at European and international level (*Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, Ministry of Higher Education and Research, Ministry for Europe and Foreign Affairs, National Institute for Research in Computer Science and Automation*).
6. Better anticipate changes in the labor market, adapt teaching methods and tools as well as the content of all higher education courses to AI, disseminate AI in all research sectors, refocus continuing education efforts, and support professional changes related to AI (*Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty,*

Ministry of Higher Education and Research, Ministry of Labor, Health, Solidarity and Families, National Institute for Research in Computer Science and Automation).

7. Support the acceleration and mass adoption by businesses over the next five years of AI use cases tailored to their needs, including by increasing communication, making available resolved use cases, supporting pioneering businesses on as yet unresolved use cases, and promoting the development of high-performance software associated with sovereign clouds (*Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, Bpifrance, National Institute for Research in Computer Science and Automation*).
8. In coordination with all stakeholders, strengthen access to data for AI systems, guarantee its quality, better protect sensitive data and intellectual property, take into account the latest scientific advances (federated learning, decentralized AI) and invest in sovereign storage capacities (*Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, National Commission for Information Technology and Civil Liberties*).
9. Build a realistic ambition for electronic components for AI and better coordinate public policy on AI with policies on the infrastructure that underpins its development, particularly with regard to electricity supply and connectivity (*Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, Bpifrance, National Institute for Research in Computer Science and Automation*).
10. Accelerate the transformation of public administrations and policies through AI, in particular through increased use of innovative public procurement and the introduction of incentive mechanisms, support measures and specific training for civil servants (*Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, National Institute for Research in Computer Science and Automation*).

Introduction

The European Regulation on Artificial Intelligence of June 2024 defines an artificial intelligence system as “*a machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments*”².

In practice, the scope of what is meant by artificial intelligence (AI) is constantly changing due to what is known as the “AI effect”: over time, certain AI techniques cease to be perceived as such as they become commonplace and the underlying technologies are mastered. For example, no one today considers a calculator to be an AI system, even though it reproduces a human cognitive function, that of counting³.

In March 2018, France launched a national strategy for artificial intelligence (*Stratégie nationale pour l’intelligence artificielle*, SNIA). The first phase of this strategy, between 2018 and 2022, focused primarily on strengthening AI research (€1.3 billion in government funding). A second, acceleration phase was announced at the end of 2021 and was implemented mainly from 2023 onwards, with the main objective of spreading artificial intelligence throughout the economy. In February 2025, on the occasion of the Paris AI Action Summit, the President of the Republic announced a third phase of the strategy, the details of which were clarified in the months that followed.

The research component of the first phase of the strategy, which in terms of political priority and budgetary effort was the main area of action in this phase, was the subject of a public policy evaluation by the Court of Auditors, published in April 2023, which this report updates and supplements⁴.

The first (Chapter I of the report) and second (Chapter II) phases of the strategy are analyzed in turn, reviewing the objectives pursued, the resources allocated, the organization of steering and implementation, the mechanisms put in place and the results achieved. The Court then analyzes the prospects for the future, identifies priorities for action and makes ten recommendations (Chapter III).

² Article 3 of Regulation (EU) 2024/1689.

³ The main technical terms and acronyms useful for understanding AI systems (LLM, GPU, HPC, GPT, GPAI, etc.) are explained in a brief glossary in Appendix 1 to this report.

⁴ Court of Auditors, *The national strategy for research into artificial intelligence: a strategy to be structured and sustained*, thematic public report, April 2023.

Chapter I

The 2018-2022 phase of the strategy: strengthening AI research

The first phase of the national AI strategy (2018-2022), funded by the government to the tune of €1.3 billion, covered only some of the issues identified at the time and required complex governance and suffered from deficient budgetary monitoring (I). This phase made it possible to begin structuring AI research and innovation ecosystem, but progress has been less clear in other areas of the strategy (II).

I - The first phase of the SNIA, in which the government invested €1.3 billion, only covered part of the issues and was based on complex governance and inadequate budgetary monitoring

Following a trend that began in several countries in the mid-2010s, France decided to launch a strategic review of the challenges associated with AI and to develop a specific public policy in this area. The objectives set for the first phase of the SNIA covered only some of the challenges identified in 2018 (A). The budget allocated by the government ultimately amounted to €1.3 billion and was poorly monitored (B). In addition, the strategy's management and implementation relied on a complex interplay between numerous actors (C).

A - The objectives set for the first phase of the SNIA covered only some of the issues identified in 2018

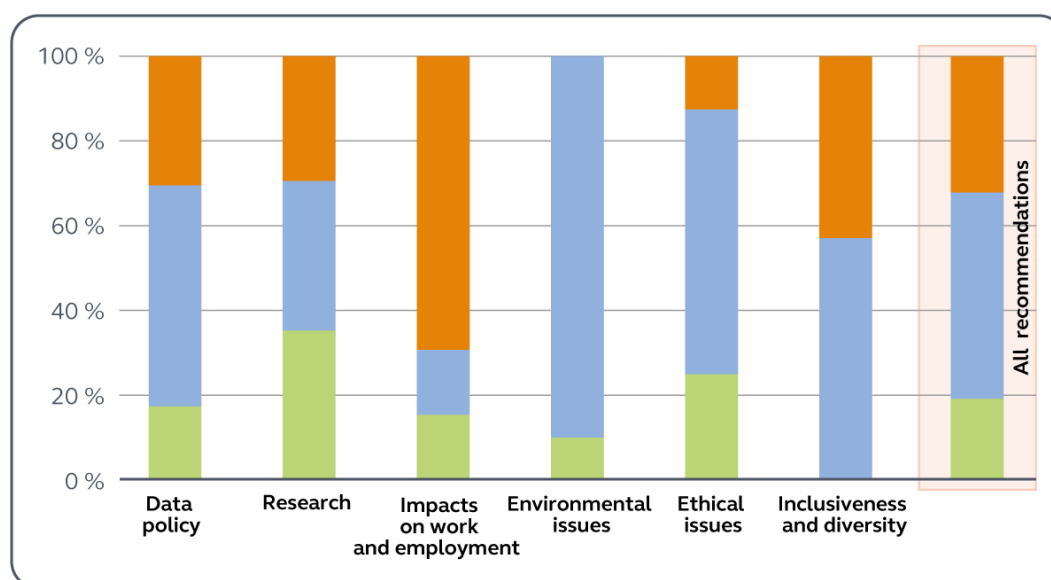
The national strategy on artificial intelligence was announced in March 2018 by the President of the Republic. This announcement followed the submission of the report by mathematician Cédric Villani⁵, which identified several critical challenges that a public policy in favor of AI would have to address. During the presidential announcement, four key areas of

⁵ Cédric Villani et al., *Giving meaning to artificial intelligence, for a national and European strategy*, March 2018.

the SNIA were outlined, covering research and higher education in AI, open data, support for priority economic sectors heavily impacted by AI, and the ethical framework for AI.

In reality, the first phase of the strategy, which ran from 2018 to 2022 (and for payments related to certain measures until 2025), focused mainly on strengthening AI research, leaving aside most of the issues related to the other areas announced in March 2018. The second phase of the strategy, between 2022 and 2025, provided an opportunity to broaden the scope of public policy in favor of AI, even though around a third of the recommendations made by the Villani mission in 2018 remain largely unimplemented in 2025 (Figure 1).

Figure n° 1 : Recommendations from the Villani report of March 2018, ranked according to their level of implementation during the first two phases of the SNIA



Key:

Green: recommendation largely committed to and implemented during the first phase 2018-2022

Blue: recommendation largely committed to and implemented during the second phase 2022-2025

Orange: recommendation still to be implemented (even if initial actions have already been taken)

Source: Court of Auditors, based on the Villani mission report of March 2018 and the national coordinator

B - The budget allocated by the government to this phase ultimately amounted to €1.3 billion and was poorly monitored

A public policy such as the SNIA is intended to draw on all the levers available to the government: regulatory power, taxation, subsidies, public-private partnerships, etc. In practice, however, it is mainly the budgetary lever that has been used.

Government funding for the first phase of the SNIA was based mainly on existing instruments, through the redeployment of ministerial budget programme appropriations and the mobilization of specific envelopes such as the Investment for the Future Programmes (*Programmes d'investissement d'avenir*, PIA) 2 and 3, the Innovation and Industry Fund, the Public Action Transformation Fund and the Skills Investment Plan. The aim was to enable the rapid announcement and deployment of the first phase without having to create new

mechanisms. In its April 2023 report, however, the Court showed that this method of financing did not guarantee the sustainability of the allocations and created a complex fiscal architecture, which made it difficult to monitor implementation. The Court’s recommendation to “*translate the public policy on artificial intelligence into a summary fiscal document that will enable its implementation to be monitored and its effects measured*” has only been partially implemented⁶, even though the budgetary framework for the second phase, financed by the France 2030 programme, has been simplified.

The amount of the budget allocated by the government to this first phase of the SNIA has fluctuated over the course of political announcements. The administrations responsible for managing the SNIA budget have not carried out rigorous, exhaustive monitoring of the appropriations that the government has planned, committed and spent for this first phase. During the audit that led to this report, these authorities had great difficulty in reconstructing consistent and reliable budgetary data on phase 1 of the SNIA. The most significant discrepancies with the figures published by the government, for example at the launch of the second phase of the strategy, resulted from difficulties in identifying the share of general funding attributable to the SNIA. The amount of private funding that contributed to the first phase of the SNIA was also subject to fluctuating announcements that were not based on a rigorous methodology. No reliable data could be established in this regard.

Subject to these reservations, it can be considered that the government, which had initially planned €1.5 billion in appropriations for the first phase of the SNIA, ultimately committed less than €1.3 billion (Table 1). AI research was the largest item of expenditure in this phase of the SNIA (nearly half of the commitments), followed by defense and security (a quarter), public transformation and the economy. The higher education component (excluding doctoral studies), which was initially planned, was not implemented, although the interdisciplinary AI institutes (3IA), without making it their priority, also carried out training activities.

⁶ See Appendix 2.

Table n° 1 : Government budget allocated to the first phase of the SNIA (2018-2022)

<i>Areas</i>	<i>Main measures</i>	Initially planned funding (in € million and as a % of the total)	Funds ultimately committed* (in € million and as a % of total)
<i>AI research</i>	3IA institutes Funding for specific AI chairs Doctoral contracts and theses Public-private partnership research Computing and data infrastructure Franco-German research partnership in AI and joint call for projects on AI for crisis response	445 29%	588 46%
<i>Defence and security</i>	ARTEMIS.IA Partnerships with research organisations research organisations	410 27%	366 27%
<i>Transformation of public action and data</i>	Call for public transformation projects Health Data Hub (Health Data Hub) AI Lab RegalIA Global Partnership for AI	154 10%	204 16%
<i>Economy</i>	Major challenges AI challenges	390 26%	114 9
<i>Higher education</i>	Non-doctoral programmes	128 8%	0 0
Total		1,527	1,272

(*) Approximations within the limits of available data

Source: Court of Auditors, based on data from the General Secretariat for Investment and the National Strategy Coordinator

C - The strategy was steered and implemented on a complex interplay between numerous stakeholders

The steering and implementation of the first phase of the strategy were complex, with more than twenty stakeholders directly involved in coordination functions (National Strategy Coordinator – *coordonnateur national de la stratégie*, CNIA –, General Secretariat for Investment – *secrétariat général pour l'investissement*, SGPI –, coordinators of related strategies, etc.), sectoral steering (National Institute for Research in Computer Science and Automation – *Institut national de recherche en informatique et en automatique*, INRIA – for research, Ministry of the Armed Forces for defense and security, Interministerial Directorate for Digital Affairs – *direction interministérielle du numérique*, DINUM – for public action transformation, Directorate-General for Enterprise – *direction générale des entreprises*, DGE – for economic and industrial aspects, Health Data Platform, etc.), operators (National Research

Agency – *Agence nationale de la recherche*, ANR –, the Public Investment Bank Bpifrance, etc.), sector monitoring (ministries responsible for the economy, higher education and research in particular) or implementation of measures (National Centre for Scientific Research – *Centre national de la recherche scientifique*, CNRS –, French Alternative Energies and Atomic Energy Institute – *Commissariat à l'énergie atomique et aux énergies alternatives*, CEA –, INRIA, 3IA institutes, etc.).

At the interministerial level, the first CNIA, between February 2018 and November 2019, was placed within a department of the Prime Minister's Office, the interministerial directorate responsible for digital technology and the State's information system, which became the DINUM in October 2019. The challenges of the industrial aspects of AI and its dissemination in the economy led to the national coordination being moved in March 2020 to the DGE of the Ministry of Economy and Finance. The coordinator heads a very small team, consisting mainly of a deputy. In June 2019, coordination of the research component was entrusted to the national research institute INRIA.

The management of the overall strategy evolved during the first phase. It was initially based on an interministerial steering committee led by the coordinator, bringing together the SGPI, nine ministries, the interministerial digital directorate, INRIA and the Health Data Hub.

France 2030 and acceleration strategies

Announced by the President of the Republic in October 2021, the France 2030 programme mobilizes €54 billion to bring about lasting change in key sectors of the French economy (energy, hydrogen, automotive, aeronautics, space) through technological innovation and industrialization. Support for strategic investment priorities can be sector-specific ("directed" innovation, €40.5 billion) or general ("structural" innovation in research, training and innovation, €13.5 billion).

In terms of "directed" innovation, some 20 national acceleration strategies have been launched with the aim of enabling the government, in collaboration with economic, social and local actors, to define its investment priorities in a number of sectors or technologies of the future by seeking to mobilize all available levers (financing, standards, research, training, etc.) in a coherent and comprehensive manner.

Three players are responsible for coordinating most of the acceleration strategies: the SGPI (quantum technologies, carbon-free hydrogen, bio-based products and sustainable fuels, emerging infectious diseases and nuclear, radiological, biological and chemical threats, cybersecurity, decarbonization of industry), the Ministry of Ecological Transition (recycling and reincorporation of recycled materials, digitalization and decarbonization of mobility, sustainable cities and innovative buildings, responsible digital technology, advanced technologies for energy systems) and the Directorate-General for Enterprise (AI, cloud, 5G and future telecommunications network technologies).

In October 2021, a strategic committee was also set up, bringing together the SGPI, the ministries most involved, INRIA, and the operators responsible for implementing the strategy, mainly the ANR and Bpifrance.

For the programmes under its responsibility, the ANR is responsible for issuing calls for projects, setting up an international jury to evaluate the quality of the projects, proposing a list of beneficiaries to the Prime Minister, signing the corresponding contracts, paying out the

grants and monitoring the execution of the contracts, and entrusting the review of the programme's results to a panel of independent experts. The process at Bpifrance follows similar steps but without calling on external experts or juries.

In April 2023, the Court showed that the SNIA's management had weakened over time. Indeed, the monitoring and consolidation of implementation indicators had gradually been abandoned, without any other performance measures being developed. The Court therefore recommended "*jointly establishing the objectives and priority indicators for public policy on AI, in line with the European strategy*". This recommendation was not implemented until 2024, as part of the rationalization of the France 2030 programme.

II - This phase enabled the structuring of AI research and innovation ecosystem, but progress has been less clear in other areas of the strategy

The creation of centers of excellence in AI research and the increase in computing capacity are among the achievements of the first phase of the SNIA, even if several limitations have emerged (A). Progress in defense and security, transforming public action and spreading AI in the economy, which were the other priorities of this phase, is less clear-cut (B).

A - The creation of centers of excellence in AI research and increased computing capacity are among the achievements of this first phase of the SNIA, even if several limitations have emerged

1 - Reminder of the conclusions of the Court's April 2023 evaluation

The most in-depth evaluation of the research component of the first phase of the SNIA is that published by the Court in April 2023. It highlighted in particular that by formalizing a strategy, the public authorities had sent a strong political signal about the importance of AI and that the priority given to research had prevented France from falling behind. The main focus of the strategy was to establish centers of excellence in AI by certifying 3IA institutes, anchored in the dynamics of integrated universities, setting up individual chairs, and then identifying centers of excellence outside the 3IA institutes. This strategy had enabled a structuring process to be initiated, whereas AI had not previously been identified as a discipline in its own right. However, the Court noted that this structuring still needed to mature, that synergies between centers of excellence needed to be strengthened, and that clarification of the missions of 3IA and "non-3IA" centers of excellence appeared necessary. This reflection had to be accompanied by a review of the timing of the funding allocated, which was based on a too short-term approach to allow for leverage effects. The follow-up to the recommendations made in this report is presented in Appendix 2.

2 - The successes of the AI research component, assessed with the benefit of two more years' hindsight

With the benefit of a further two years' hindsight, an assessment of the results of the AI research component of the first phase of the strategy highlights several successes.

(a) A first success is the beginning of the structuring of centers of excellence in AI research and the establishment of a “pact of trust” with the universities concerned, in order to place the objectives of the SNIA within a strategic framework for a given territory and to move beyond the opportunistic logic of seeking additional research funding.

The four 3IA institutes, selected by an international jury in April 2019 and approved in the summer of 2020, bring together 150 chairs, 8% of which are international. These are PR[AI]RIE, initially led by INRIA (48 chairs), 3IA Côte d’Azur, led by Université Côte d’Azur (47 chairs), MIAI, led by Université Grenoble Alpes (31 chairs) and ANITI, led by Université fédérale Toulouse Midi-Pyrénées (24 chairs). In each of these clusters, the university plays a leading role, as it is the only institution capable of integrating the three components of research, higher education and innovation, in close collaboration with academic, entrepreneurial and industrial players in the region.

These centers have gradually begun to develop interdisciplinary, cutting-edge AI research based on the existing foundation, in line with leading global research, and to build links between fundamental AI research, areas of integration and sectors of application. According to INRIA estimates, in four years, this network and its more than 150 partners, mainly companies, have brought together more than 500 researchers, funded nearly 500 doctoral and post-doctoral students, and trained more than 13,000 people each year. The 3IA networks have also entered into more than a hundred industrial partnerships with leading companies.

Forty-three additional chairs in artificial intelligence, four of which are co-funded by the Defense Innovation Agency, were also established in 2020. These have largely been integrated into structural initiatives such as DATAIA (Paris Saclay University), Hi! PARIS (Paris Institute of Technology and HEC Paris) and SCAI (Sorbonne University Alliance), which have subsequently become “AI Clusters”. These chairs are co-funded by institutions, companies and local authorities.

Around 300 doctoral contracts have also been signed, as well as around 100 theses under industrial research training agreements.

The importance and visibility of artificial intelligence in the three major national research institutes, CEA, CNRS, with the creation of the “AI for Science and Science for AI” (AISSAI) center in particular, and INRIA, has grown well beyond SNIA funding.

A virtuous circle has also been set in motion as regards partnership research programs in the field of AI. Research expenditure to accelerate innovation in companies undertaken by the Carnot network research institutes and technological research institutes between 2018 and 2022 (approximately €160 million, almost exclusively in AI) was four times higher than initial forecasts in 2018 and covered a variety of fields. Outside the SNIA, national research institutes such as INRIA have also developed several instruments for partnership-based research.

(b) The second significant success of the first phase of the SNIA was the opening up of structural computing facilities, which created a favorable environment for AI research.

The Jean Zay supercomputer, operated by the CNRS on behalf of the French National High-Performance Computing Centre (*Grand équipement national de calcul intensif*, GENCI), has been operational since 2019 on the Paris-Saclay University campus. When it was commissioned, it offered a computing power of 16 petaflops/s, which placed it far behind other AI computing infrastructures of the time installed in the United States (e.g. the Summit and Sierra supercomputers, with a power of 200 and 125 petaflops/s respectively) or even in Europe (e.g. the Swiss Piz Daint or Italian Marconi 100 supercomputers, with 25 and 22 petaflops/s respectively). During the first phase of the SNIA, extensions were gradually implemented in March and November 2020 and then in June 2022 (bringing the computing power of the Jean Zay supercomputer to 37 petaflops) to meet growing demand. With each extension, demand exceeded the availability of the machines.

Beyond the implementation of the infrastructure, as part of the national AI research programme (PNRIA), a network of engineers was set up at Jean Zay and in four regional centers, available to academic and industrial users of the supercomputer. The link between computing infrastructure, research and innovation, with projects with strong academic and industrial roots, and the support provided by a dedicated team of engineers, was a source of inspiration for the EuroHPC/AI Factories programmes subsequently developed by the European Commission.

The supercomputer has enabled numerous research teams, companies and administrations to access computing power to develop AI systems at reduced or even zero cost in the case of open research, which is the vast majority of cases.

In addition, Jean Zay is one of the most environmentally friendly supercomputers in the world, thanks to its internal cooling system and its mechanism for recovering waste heat from the machine. This recovery makes it possible to heat the equivalent of 1,000 homes (i.e. 4 GWh per year, compared to an power consumption of 19 GWh per year) on the Saclay plateau, a first in Europe on this scale.

(c) Third success of the first phase of the SNIA, the position of French research in AI has been strengthened at European level and in terms of international cooperation, with a focus on attractiveness issues.

French AI research institutes are heavily involved, for example, in the European programmes Tailor, a network of research centers on the foundations of trustworthy AI, HumanE-AI-Net on AI systems that complement human intelligence, AI4Media on AI for media and society, Ellis on European excellence in learning and intelligent systems, and the Adra partnership to facilitate synergies between the AI, data and robotics communities.

Franco-German cooperation in AI research has been strengthened with the establishment of a strategic roadmap, followed by calls for joint projects to strengthen links between French and German research institutes and support applied research projects by innovative emerging players in AI on risk management, and the establishment of a strategic framework agreement between INRIA and the German AI research center in January 2020.

Finally, particular attention has been paid to the international attractiveness of French AI research, for example at the CNRS and INRIA, with the aim of first preventing the brain drain and, if possible, bringing back or attracting talent from abroad.

3 - The limitations of the AI research component

The implementation of the AI research component of the first phase of the national strategy also encountered several limitations.

(a) The structural effect of the measures funded was not always promoted.

The government's use of a series of calls for projects or expressions of interest for one-off funding with a limited lifespan presents the risk of increasing fragmentation within research, to the detriment of overall consistency, the creation of critical mass and more structural strategies enabling coordination, complementarity and accountability among stakeholders over the long term.

The selection process for 3IA institutes, based on an international jury, focused less on the structural qualities of the actors than on the intrinsic characteristics of the projects presented (unlike the analysis that a private investment fund, for example, would have conducted). This may have led to the exclusion of major academic players such as Paris-Saclay University and to the adoption of orientations which, given the rapid evolution of technologies and the slow start-up of the institutes, only from 2021 onwards, were likely to become obsolete. The long-term funding of these institutes remained uncertain until they were transformed, after a lengthy and erratic process, into "AI Clusters" as part of the second phase of the SNIA.

Furthermore, the lack of alignment among stakeholders on both the priorities set by the government and how to implement them may have reduced the effectiveness of the first phase of the strategy.

Finally, the difficulty in articulating national and European strategies, going beyond a funding approach, was an obstacle to a long-term vision.

(b) The "higher education" component, initially planned for the first phase of the SNIA, was not implemented. More broadly, the issue of adapting school education and all initial and continuing training to artificial intelligence, and that of raising awareness of the new ethical issues raised by this technology, were not priorities in this phase of the national strategy. For example, the government's cancellation of the "digital and education" priority research programme and equipment (*programme et équipement prioritaire de recherche*, PEPR) (€77 million planned), after more than two years of work and despite an announcement of its launch by the Prime Minister in October 2021, has set back the implementation of a structured transformation of learning through AI and the construction and deployment of sovereign building blocks of artificial intelligence in schools and higher education by several years.

(c) The attractiveness of AI research talent from abroad, particularly through the "Choose France" scheme, has not lived up to expectations. Among the obstacles mentioned by some stakeholders are the lack of attractive salaries and the absence of support measures for settling in, in a rapidly developing and particularly competitive sector.

(d) The difficulty of moving from a reactive to a proactive approach is also mentioned, particularly at the time of the generative AI shock. From this point of view, the BigScience project was a scientific success, with the development of the Bloom language model, whose performance was similar to that of its contemporary GPT3, but it could not be transformed into an innovation comparable to ChatGPT, due to the lack of strong governance to monitor the project and mobilize a consortium of companies and funding commensurate with the challenge.

B - Progress in defense and security, transforming public action and spreading AI in the economy, which were the other priorities of this phase, is less clear-cut

The results of the first phase of the SNIA on the defense and security, transformation of public action and businesses components have not been evaluated in a manner comparable to that published by the Court in April 2023 on AI research.

1 - Defense and security

The “defense and security” component of the first phase of the SNIA (€366 million in committed funds) was not actually functionally integrated into the strategy but remained under the management of the Ministry of the Armed Forces. During the period covered by the 2019-2025 military programming law, this ministry devoted an average of €100 million per year to AI projects.

The ARTEMIS.IA project, led by the Digital Defense Agency, aims to provide the Ministry with a sovereign and secure solution for mass data processing and AI. The initial phases of the project validated the technical solutions selected and brought together academic and industrial players. The implementation phase, launched in 2022, enabled the deployment of an initial operational platform from 2023 onwards for the joint intelligence function.

Structural partnerships with research institutes (CNRS, INRIA, CEA) have been established to develop dual-use projects, particularly those with high potential for technological spin-offs for defense. INRIA began by setting up an internal structure, *INRIA Défense*, in March 2020, which was subsequently placed under joint governance with the Ministry and has carried out more than 70 projects, almost all of which relate to AI.

A defense ethics committee was set up in January 2020 to address issues raised by emerging technologies, including AI, in the field of defense. It has issued opinions on the integration of autonomy into lethal weapons systems (April 2021), the digital environment of combatants (April 2022) and the use of AI technologies by the armed forces (January 2025).

2 - Transformation of public action and data

The “transformation of public action and data” component of the first phase of the SNIA was led by the Interministerial Directorate for Digital and State Information Systems (which became DINUM in October 2019). More than €200 million in funding was committed to this component, mainly through the Fund for the Transformation of Public Action.

Two initiatives accounted for most of the funding, with around €80 million each: the call for public transformation projects (a) and the establishment of the health data hub (b). Other AI-driven public transformation projects during this period include the Lab IA (c), RegalIA (d) and the Global Partnership for AI (e).

(a) The call for public transformation projects supported 46 AI-related projects, with funding for “entrepreneurs of general interest” to meet the needs expressed by government departments. Funding was provided for projects aimed at improving the detection of car models and reducing errors in the issuing of traffic fines, anticipating the difficulties faced by

businesses in order to better support them, pseudonymizing court decisions and detecting fake reviews on the internet. However, as interesting as they were, these projects remained isolated initiatives and their transformative effect on public action was very limited.

(b) The *Health Data Hub* platform aims to bring together all the health data of the population receiving care in France and to guarantee simplified, unified, transparent and secure access to this data, in particular to promote medical research through artificial intelligence systems. The platform, established in July 2019 as a “public interest group” and whose missions are set out in Article L. 1462-1 of the Public Health Code, is responsible for implementing the strategic guidelines set by the government for the national health data system (*système national des données de santé*, SNDS). These missions include supporting health data managers in the collection, consolidation and enhancement of their assets, facilitating the use of health data with simplified and accelerated access for all project leaders, protecting citizens’ health data, and innovating alongside research and industry players to develop the best solutions for patients.

In the context of the health crisis, the platform was brought into service ahead of schedule, in April 2020, on a limited scale. The decision to host the data on the cloud of a large American company has led to heated controversy over issues of sovereignty and personal data protection. In an order dated 13 October 2020, the Council of State observed that this data could not legally be transferred outside the European Union (EU) and that the risk of US intelligence services requesting access to it could not be completely ruled out. This situation required special precautions to be taken, under the supervision of the French Data Protection Authority (*Commission nationale de l’informatique et des libertés*, CNIL).

While emphasizing the difficulty of finding effective sovereign alternatives, the Government committed at the end of 2020 to migrate the platform to a European solution within two years. However, this migration was not completed within the announced timeframe. Nor was the platform affected by the doctrine gradually developed by the government, applicable to new projects, of “trusted cloud” in 2021, then “cloud at the center” in May 2023, involving a high level of cybersecurity certification (*SecNumCloud*).

In this context, the platform has only been operating at a slow pace. Anticipating a negative response, in January 2022 the government decided not to ask the CNIL for authorization to have a copy of the main SNDS databases. In addition, research or innovation projects that plan to use French health data, including for training AI systems, are subject to a case-by-case prior authorization procedure by the CNIL and a scientific and ethical committee. Combined with the processing times of the National Health Insurance Fund (*Caisse nationale d’assurance maladie*, CNAM), which is currently performing the tasks of the Health Data Hub without any new resources, the implementation of this procedure can take more than six months. This delay is often considered excessive by applicants.

Despite the efforts made by the CNIA and the DGE’s study to analyze French cloud offerings, this file has remained focused on the question of sovereign data hosting, without considering the economic and public health benefits that could result from the use of this data by AI systems alongside the actual risks involved, nor comparing the situation to other equally sensitive cases of data storage on non-sovereign clouds.

In its December 2023 report, the mission led by Jérôme Marchand-Arvier, State Councilor, tasked by the Government with laying the foundations for a roadmap for the reuse

of health data⁷, concluded that the significant potential for reuse of France's rich health data heritage remained under-exploited, in a competitive international context, linked in particular to the development of AI, that there were still many obstacles to the reuse of health data and that, despite the investment of the Health Data Hub, the sector was marked by a lack of cooperation and trust. The mission formulated some forty proposals, grouped into five areas, including a prerequisite: to arbitrate the issue of hosting a copy of the main SNDS database on the platform.

In the chapter of its May 2024 report on the application of Social Security financing laws devoted to the SNDS⁸, the Court of Auditors also emphasized that the results achieved had fallen far short of the objectives set by the legislator and made recommendations aimed at enabling effective, faster and more extensive access to health data.

In July 2025, several measures were announced in an attempt to finally overcome these difficulties:

- concerning the secondary use of health data, creation of a forum for dialogue on the use of health data (involving healthcare establishments, healthcare professionals, public research, industry and industrial sectors, patients and users), launch of an "AI and health data" strategy to better organize, protect and promote the secondary use of this data for research, innovation and public policy⁹, and launch of a public contract for the sovereign hosting of a copy of the main SNDS database and the decongestion of current processing capacities (commissioning announced for summer 2026);
- concerning the further exploration of issues related to AI in healthcare, organization of a public consultation around four areas (clarifying regulations and framing the uses of AI in healthcare; strengthening the evaluation of AI solutions and their impact on the healthcare system; supporting professionals and institutions in their adoption of the tools; building a sustainable economic framework to support AI-based innovation) and launching two calls for expressions of interest to test AI solutions in healthcare and medical-social institutions (on emergency medicine and working hours in health institutions).

(c) The Lab IA supported 25 projects to test AI in government departments, developed shared resources for AI within the government and led an "AI and data science" community.

(d) The RegalIA pilot project, led by INRIA, mobilized public research in support of the missions of the Digital Regulation Expertise Centre (*Pôle d'expertise de la régulation numérique*, PEReN), attached to the DGE, responsible for analyzing the functioning of digital platforms to enable the competent authorities to adapt or implement a regulatory framework.

(e) The Global Partnership on AI (GPAI), launched on the initiative of Canada and France during their G7 presidencies in 2018 and 2019, aims to support international expertise to guide public authorities on the development of trustworthy AI, the impact of AI on work, data governance, and the innovation and trading of AI-related technologies.

⁷ Jérôme Marchand-Arvier et al., *Bringing together stakeholders in the ecosystem to enable the secondary use of health data*, 5 December 2023 ([rapport donnees_de_sante.pdf](#)).

⁸ Court of Auditors, *Report on the application of Social Security financing laws 2024, chapter on the national health data system, May 2024*.

⁹ *Artificial intelligence and health data strategy*.

France has also been heavily involved in the work carried out by the Organization for Economic Co-operation and Development (OECD) to define in 2019 and then regularly update the “Principles on AI”¹⁰, the first intergovernmental standard promoting innovative and trustworthy AI.

3 - Economy

The economic component of the first phase of the SNIA was presented in July 2019, with objectives and financial resources that have fluctuated significantly over time. Ultimately, the commitments that can be linked to the strategy can be estimated at more than €110 million.

The successive coordinators of the SNIA, as well as operators (notably Bpifrance) and various strategy leaders (notably INRIA) have played an essential role in consolidating and animating a French AI “ecosystem”, where centers of research excellence, companies, innovators, startups and financiers interact. In the opinion of many stakeholders and observers, this “intangible asset”, which was built up gradually and accelerated significantly during the second phase of the strategy, has been one of the SNIA’s main added values.

At its inaugural meeting in July 2018, the Innovation Council selected two “major challenges” in AI to be funded by the Innovation and Industry Fund: *“How can medical diagnostics be improved through artificial intelligence?”* and *“How can we secure, make reliable and certify critical artificial intelligence systems?”* The momentum created around these “major challenges” waned when the people responsible for coordinating them left, leading to inadequate monitoring of funding and results.

The first of these “major challenges” aimed to accelerate the development of new products and promote the introduction of individualized patient care. The results of this major challenge, which funded 17 projects, are mixed, particularly for projects involving competitiveness clusters. Several of them had not yet been completed by June 2025. The main project, led by the federation of medical specialties, had a very low implementation rate.

The second major challenge (*“How can we secure, make reliable and certify critical artificial intelligence systems?”*) funded three series of actions: Confiance.ai, a platform of software tools for industry, which enabled the online publication of more than a hundred software components for trustworthy AI at various levels of maturity and the publication in October 2022 of a first white paper on trustworthy AI in critical systems¹¹, the Research and Investment Platform for the Safety and Security of Autonomous Mobility (*Plateforme de recherche et d’investissement pour la sûreté et la sécurité de la mobilité autonome*, PRISSMA), and the AI standardization strategy coordinated by the French Standardization Association (AFNOR), which aims to involve a large number of companies in European and international AI standardization bodies.

As part of the structural industrial challenges supported by Bpifrance, more than 600 projects integrating AI that are key to competitiveness have been supported since 2020, particularly in the health and automotive sectors.

¹⁰www.oecd.org/fr/themes/principes-de-l-ia.html .

¹¹ [White Paper-Confiance.ai-October2022-1.pdf](#).

A dozen “AI Challenges”, funded by PIA funds, have enabled support for ten companies and seven public administrations in the four sectors identified as priorities by the Villani mission (defense and security, health, environment, transport and mobility). However, an evaluation by the DGE showed that this programme was not very effective, as sponsors only adopted the technologies in about one in ten cases, in the absence of a rigorous assessment of the robustness of the AI solutions developed for a given use case.

A Manifesto of sixteen industrial groups¹² was also drawn up to support a dozen research and teaching chairs and fund more than a hundred theses in AI.

For their part, the 3IA institutes also proposed six projects to create AI startups, half of which have been launched.

Although not funded by the SNIA, several INRIA initiatives focus largely on AI: integrated research projects with industry (12 over the period 2019-2023, with a target of 30 over the period 2024-2028), the “INRIA challenges”, which focus on broader partnership research topics involving several teams, the creation of startups with the INRIA Startup Studio, a support mechanism for technological entrepreneurship arising from public research, intensive in software, i.e. mainly related to AI (over the period 2020-2024, some 600 projects have been identified, 160 supported, and more than 60 startups created), and LaborIA, a joint laboratory between the Ministry of Labor and INRIA set up to study the impacts of AI deployment on the world of work, activities, workers, social dialogue, skills and training needs.

CONCLUSION

The implementation of the first phase of the national strategy for artificial intelligence has made it possible to initiate a public policy on AI in France, even if it has only been able to cover part of the issues identified in March 2018. The steering and implementation of this first phase of the strategy was based on a complex interplay between numerous actors. The diverse resources allocated by the government ultimately amounted to €1.3 billion, and their monitoring proved to be deficient.

Despite several limitations, the main contribution of this 2018-2022 phase is that it helped to initiate the development and structuring of research and innovation in the field of AI, with the creation of centers of excellence, the opening of essential computing infrastructures and support for the growth of AI start-ups in a variety of fields.

In other areas covered by the SNIA – defense and security, the transformation of public action and the dissemination of artificial intelligence in the economy – progress has been less clear-cut. Several of the announced priorities, particularly those relating to the key issues of training and supporting change in the economic sectors most affected by artificial intelligence, have not been implemented, or only to a very limited extent, at the risk of causing France to fall behind.

¹² Airbus, Air Liquide, Dassault Aviation, EDF, Michelin, Orange, Naval Group, Renault, Safran, Saint Gobain, Schlumberger, STMicroelectronics, Thales, Total, Valeo and Veolia.

Chapter II

The 2023-2025 phase of the strategy: aiming for the diffusion of AI in the economy

The 2023-2025 phase of the national strategy, in which the government has invested €1.1 billion, aims to promote the diffusion of AI technologies in the economy (I). The structuring of centers of excellence in AI research and higher education has been consolidated (II). France has achieved initial successes in generative AI and in the areas of frugality and trust (III). In addition, it has begun to play an active role in AI on the European and international stages (IV). However, the challenge of massification and support for the spread of AI beyond the circle of specialists – businesses, public administrations, students, citizens – has not yet received sufficient attention (V).

I - The second phase, in which the government invested €1.1 billion, aimed to promote the diffusion of AI in the economy

Launched without prior evaluation, the second phase of the SNIA, focused on the dissemination of artificial intelligence technologies in the economy, saw its priorities, budget and timetable change significantly from the initial announcements (A). Despite the ambition to accelerate progress, the slow start-up of most of the measures resulted in a low level of budget consumption (B). The governance of the national strategy remained complex, despite some promising developments and examples of successful coordination (C).

A - Launched without prior evaluation, this phase, focused on the dissemination of AI technologies in the economy, has seen its priorities, budget and timetable changed significantly from the initial announcements

A second phase of the SNIA was announced by the Government in November 2021. This phase, known as the acceleration phase, among other strategies to accelerate the France 2030 programme, was initially presented under the title “*Attracting talent and transforming our scientific potential into economic success*”, with three “*key objectives*” (higher education and

research in AI, embedded, sober and trustworthy AI, diffusion of AI) associated with “*success indicators for 2025*”. In financial terms, the stated ambition was for a total budget of €2.1 billion, of which €1.5 billion would be financed by the government as part of PIA 4, which was subsequently integrated into the France 2030 programme.

The launch of this second phase was not preceded by any overall assessment of the actions undertaken and the results achieved during the first phase. The first in-progress evaluation of France 2030 by the Future Investments Supervisory Committee, published in June 2023, did not seek to retrospectively assess the actions undertaken during the first phase of the SNIA.

In reality, given the substantial changes that this phase of the strategy has undergone, the framework announced in November 2021 could not be fully respected in terms of either its objectives or its budget. A degree of flexibility was applied when setting the objectives, given the fiscal constraints (redeployments in 2023 within the France 2030 programme) and the rapid evolution of AI-related technologies. In particular, it became necessary to adjust the strategy due to the breakthrough of generative AI and large language models, an area in which France’s technological dependence appeared to be almost complete at the time. In return, and in a targeted manner, certain measures deemed inconclusive were discontinued and others saw their funding reduced, notably the “AI Clusters” call for expressions of interest (CEI), whose allocated funding was reduced from €450 million to €360 million and whose contract duration was reduced from seven to five years.

Finally, although it was not clearly formalized, the guiding principle of this second phase in its new format was considered to be the diffusion of AI in the economy, and it was structured around four priorities, with support for generative AI being added to the objectives defined at the end of 2021. Four overarching monitoring indicators were selected in 2024 and accompanied by targets for 2025-2030 (Table 2). These indicators, used for communicating on results, are updated annually, which limits their use at the operational level. At the national coordinator level, the strategy was steered more by consultation with high-level experts than by indicators, which would have risked becoming quickly outdated. In practice, operational monitoring of this phase of the strategy focused on industrial policy in favor of actors in the AI value chain across the four strategic areas and the positioning of training and research in line with these areas.

The France 2030 budget allocated to this phase of the strategy ultimately amounted to €1.055 billion, which is one-third less than what had been announced in November 2021. Support for higher education and research in AI accounts for two-thirds of the funding. However, the share of funding devoted to supporting the integration of AI solutions into companies’ business models and operational processes remained very low.

**Table n° 2 : Priorities, indicators and budget finally adopted
for the second phase of the SNIA**

<i>Priorities</i>	<i>Measures*</i>	<i>Indicators (targets 2025 → 2030)</i>	<i>Planned budget for France 2030 (in € million and as a % of total)</i>
<i>Higher education and research in AI</i>	AI Clusters Skills and Jobs of the Future PEPR AI AI Cluster Chairs (AI Faculty France) P16 (Scikit-Learn) Maturation-pre- maturation	Total number of people with AI degrees or certifications through programmes funded under the SNIA (50,000 → 400,000) Global ranking of R&D institutions in AI, particularly in terms of publications** (→ France 1 st EU country)	687.9 65%
<i>Generative AI</i>	Digital commons for generative AI Jean Zay calculator extension Acceleration of generative AI applications PNRIA network of engineers ALT-EDIC	Total funding raised by French AI start-ups (→ France 1 st EU country)	136.2 13%
<i>Embedded, sober and trustworthy AI</i>	Embedded AI Electronic IPCEI and connectivity INESIA Edge AI DIAT DeepGreen Sectoral TEF	(same indicator as for generative AI)	221.0 21%
<i>AI adoption by companies</i>	AI Booster	Percentage of companies with more than 10 employees using an AI solution in their processes: - specific (7% → 20%) - off-the-shelf, without adjustment to the company's use case (15% → 40%)	10.0 1%
Total			1,055.2

(*) Schemes ranked in descending order of budget. Acronyms are explained below (Parts II to V)

(**) Global Vibrancy Ranking and Global AI Index

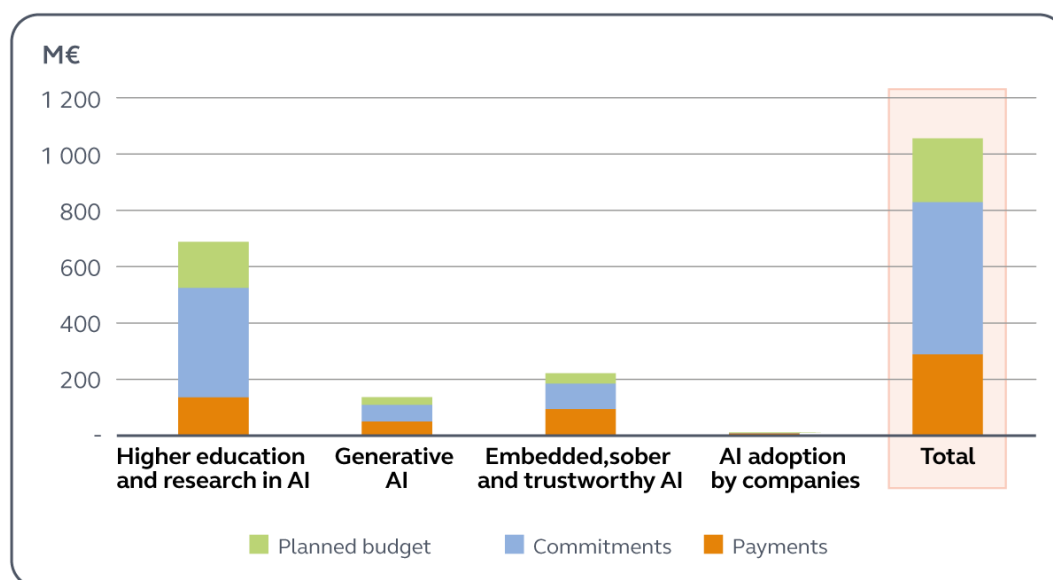
Source: Court of Auditors, based on data from the National AI Coordinator

B - The slow start-up of most measures resulted in low levels of budget consumption

Compared to the first phase of the strategy, the financing of the second phase, based entirely on appropriations from the France 2030 programme, simplified its budgetary governance and facilitated the monitoring of its implementation. The General Secretariat for Investment (SGPI) also acquired enhanced tools in 2024 to improve the monitoring of budgetary and extra-budgetary flows, and submits quarterly financial reports to Parliament. At the time of the Court's audit, however, difficulties remained in reconciling data between the various actors.

As at 30 June 2025, commitments amounted to €830 million, or 79% of the planned budget, and payments to €289 million, or 35% of commitments (Figure 2).

Figure n° 2 : Budget implementation for the second phase of the SNIA



Commitments and payments as at 30 June 2025

Source: Court of Auditors, based on data from the SGPI and the National Coordinator for AI

This low rate of consumption, at a time when a third phase of the SNIA has already been announced, reflects the late start, from 2023, of most of the instruments in this “acceleration” phase, which only became effective one to three years after the initial announcements. In particular, the first commitments of funds for the flagship “AI Clusters” scheme (which has the largest budget in the strategy) were not finalized until the end of 2024, three years after they were announced and two years after the candidates submitted their bids.

C - Governance of the national strategy remained complex, despite some promising developments and examples of successful coordination.

1 - Overview

In the composition of the Government, the priority given to AI was reflected for the first time in September 2024 in the title of “Secretary of State for Artificial Intelligence and Digital Technology” (and no longer just Digital Technology), attached on a temporary basis to the Minister for Higher Education and Research. In December 2024, they became the “Minister Delegate for Artificial Intelligence and Digital Technology” and was attached to the Minister for the Economy, Finance and Industrial and Energy Sovereignty.

The national AI coordinator (CNIA), who heads a very small team (essentially one deputy), has remained attached to the Directorate-General for Enterprise (DGE), and the comitology for steering the strategy has changed little since the first phase: the steering committee (whose membership has been expanded) and the monthly strategic committee, chaired by the CNIA, ensure collegial operational management of the strategy, monitor its implementation and guarantee its overall consistency; the strategic committee prepares the administrations’ positions prior to the monthly meetings of the ministerial operational digital steering committee and, once approved, the projects are subject to a formal investment decision signed by the SGPI; investment decisions exceeding €15 million must be approved by the France 2030 executive committee, which brings together the relevant central administration directors, and receive the Prime Minister’s approval, a process that takes several months.

The coordination mission entrusted to INRIA for the research component of the SNIA has been renewed and extended to the “higher education in AI” component by a mission letter dated April 2024. Within INRIA, the digital programme agency¹³, which was first proposed in 2021 and created in January 2024, aims to strengthen the collective dynamics of higher education and research by bringing together the various stakeholders and increasing public-private dynamics.

2 - Coordination with other acceleration strategies and priority initiatives of France 2030

Coordination between the SNIA and other acceleration strategies and priority initiatives of France 2030 relating to digital technology and related fields takes place at several levels: the coordinators of the acceleration strategies on electronics, cloud HPC and digital health, as well as the National Cybersecurity Agency (ANSSI), the SGPI, the DINUM and the Interministerial Directorate for Public Transformation (DITP), participate in the SNIA steering committee; conversely, the CNIA participates in the digital health strategy steering committee; Coordination also takes place at the level of the ministerial operational steering committee for digital technology, which is the interministerial governance body for France 2030 on digital issues.

¹³ Court of Auditors, *Programme Agencies*, communication to the Finance Committee of the National Assembly (to be published in 2025).

These coordination efforts have made it possible, for example, to develop the call for projects on embedded AI in conjunction with the SNIA and the acceleration strategy on electronics, also led by the DGE, by integrating requirements in terms of components, architecture and usage constraints. Funding for the extension of the Jean Zay supercomputer and the Alice Recoque exascale supercomputer project was coordinated with the HPC cloud strategy. The AI component is becoming increasingly prominent in calls for projects under other digital acceleration strategies, confirming that AI is a general-purpose technology but also reflecting a gradual convergence of strategies, particularly on the issues of mass data processing, optimization and technological interfaces.

In addition, an “AI Convergence” plan, led by the SGPI in conjunction with the CNIA, aims to strengthen the link between the SNIA and all of France 2030’s acceleration strategies. It has been estimated that, as of 31 December 2024, a total of €3.4 billion had been committed to AI within France 2030. Several SGPI investments contribute in particular to the development of “enabling” technologies for AI, such as the cloud, data storage and structuring, energy supply (particularly nuclear), electronics, 5G and future 6G communication networks, etc. However, the “Convergence IA” initiative does not constitute a comprehensive public policy on AI, nor does it accelerate the integration of AI into all innovation policies.

II - The structuring of centers of excellence in AI research and higher education has been consolidated

The implementation of the second phase of the SNIA has enabled the development of several instruments to strengthen the structuring and excellence of research and higher education in artificial intelligence (A). France’s position in AI research and higher education continues to progress, even if several limitations remain (B).

A - The second phase of the SNIA has enabled the development of several instruments to strengthen the structuring and excellence of research and higher education in AI

1 - Overview

The AI research and higher education component is a continuation of the first phase of the strategy concerning research, with a focus on expansion and acceleration, particularly in higher education. Two-thirds of the final budget allocated to the strategy is devoted to this area, representing €688 million. More than 80% of this budget is concentrated on three flagship initiatives: the calls for expressions of interest “AI Clusters” (€360 million) and “Skills and Jobs of the Future” (*Compétences et métiers d’avenir*, CMA, €200 million), and the priority programme and equipment for AI research (€73 million). As of 30 June 2025, the funds committed to this “AI research and higher education” component amounted to €525 million, or 76% of the planned budget, and payments amounted to €136 million, or 26% of the funds committed (Table 3).

**Table n° 3 : “AI research and higher education” component
of the second phase of the SNIA**

Measures	Content	Start-up (initial announceme nt of the scheme / first funding commitments)	France 2030 planned budget (in € million and as a % of the total)	Commitment s of funds as at 30 April 2025 (in € million)	Payments as at 30 June 2025 (in € million)
AI Clusters <i>Operator ANR, steering INRIA, also monitored by DGESIP and DGRI</i>	9 clusters of excellence selected in research and higher education in AI	November 2021 / November 2024	360.0 52%	360.0	61.2
Skills and Jobs of the Future <i>Operator ANR, monitoring DGESIP</i>	Adaptation of training programmes to skills requirements.	December 2021 / June 2022	200.0 29%	89.0	43.7
PEPR IA <i>Operator ANR, steering INRIA, CNRS and CEA , also monitored by DGRI</i>	Support for upstream research to promote disruptive innovation breakthrough innovation	November 2021 / October 2023	73.0 11%	61.5	25.0
AI Cluster Chairs (AI Faculty France) <i>Operator not yet designated, steering INRIA, also monitored by DGESIP</i>	Funding of research chairs subject to the creation of permanent positions	May 2024 / referred to the third phase of the SNIA	40.0 6%	0	0
P16 (Scikit-Learn) <i>Operator ANR, steering INRIA, also monitored by DGE and DGRI</i>	Development of software libraries, in particular Scikit-Learn, with promotion by the company Probabl	November 2021 / February 2024	7.9 1%	7.9	2.9
Maturation-pre- maturation <i>Operator ANR, steering CNIA, also monitored by DGE</i>	Co-financing of projects for business creation via technology transfer acceleration	November 2021 / February 2023	7.0 1%	7.0	3.5

<i>Measures</i>	<i>Content</i>	Start-up (initial announceme nt of the scheme / first funding commitments)	France 2030 planned budget (in € million and as a % of the total)	Commitment s of funds as at 30 April 2025 (in € million)	Payments as at 30 June 2025 (in € million)
	companies				
Total			687.9	525.4	136.3
<i>Committed / Planned</i>				76%	
<i>Consumed / Committed</i>					26%

Source: Court of Auditors, based on SGPI and CNIA

2 - AI Clusters

The call for projects “AI Clusters” (€360 million planned and committed, €61 million disbursed as of 30 June 2025) aims to support the emergence of centers of excellence in research and higher education in the field of AI. Drawing on the expertise of an international jury, the government selected nine clusters, four of which had already been supported during the first phase of the strategy as “3IA” institutes, and three others that had been structured for several years outside of this certification and had received funding from the national strategy for numerous individual AI research chairs (Table 4).

Table n° 4 : “AI Clusters”, centers of excellence in AI research and training

<i>AI Cluster</i>	<i>University</i>	<i>Background</i>	<i>Specialities</i>	Budget allocated to France 2030	Private co- financing
<i>PR[AI]RIE - PSAI (Paris School of AI)</i>	Paris Sciences et Lettres University	3IA Institute	Health, society, digital humanities	€75 million	33% (Facebook France, GE Healthcare)
<i>MIAI Cluster</i>	Grenoble Alpes University	3IA Institute	Systems, energy, industry	€70 million	36% (Eviden, Schneider, Huawei, Enedis, Naverlabs, ST Microelectronics, Braincube, Pelican Health)
<i>Hi! PARIS Cluster 2030</i>	Institut Polytechnique de Paris and HEC Paris	Structuring outside 3IA	Economics, management and business, engineering	€70 million	43% (CapGemini, Total Energies, Schneider Electric, L’Oréal, Vinci)
<i>PostGenAI@ PARIS</i>	Sorbonne University	Structuring outside 3IA	Law, society and ethics, biosciences and health	€35 million	42% (Altair, Thales, Matrice, Suez, Qubit Pharmaceuticals, Meta)

<i>AI Cluster</i>	<i>University</i>	<i>Background</i>	<i>Specialities</i>	<i>Budget allocated to France 2030</i>	<i>Private co-financing</i>
ENACT	Universities of Lorraine and Strasbourg	-	Human-AI interaction, materials, health	€30 million	48% (VASP, Safran Defence & Space, AstraZeneca, Medtronic, Intuitive, In Simo, Equasens)
DATAIA	Paris Saclay University	Structuring outside 3IA	Physics, health (oncology)	€20 million	220% (RTE, Servier, GE Healthcare, Safran, Michelin, EDF, SLB, Magentine)
ANITI AI	University of Toulouse	3IA Institute	Aeronautics, transport	€20 million	20% (Airbus, Alstom, RTE, Renault)
3IA Côte d’Azur 2030	University Côte d’Azur	3IA Institute	Health, observation and environment	€20 million	222% (Amadeus, GDG, NXP, SAP)
Sequoia	University of Rennes	-	Cybersecurity, environment	€20 million	45% (Thales, Airbus, Safran, Neverhack)

Source: Court of Auditors, based on ANR data

The target in terms of training volume for these “AI Clusters” is to reach more than 100,000 AI graduates per year by 2030, at bachelor’s, master’s, doctoral and continuing education levels, compared to less than 40,000 in 2023, the reference year in the CEI dossier (Table 5). The ANR’s evaluation process for AI Clusters provides for the first results to be measured in 2026.

Table n° 5 : Targets for the number of people trained in the “AI Clusters”

<i>Training courses in “Clusters”</i>	<i>2023 (achieved)</i>	<i>2030 (target)</i>
<i>Bachelor’s degree</i>	9,000	34,000
<i>Master</i>	21,000	45,000
<i>Doctorate</i>	2,200	3,600
<i>Continuing education</i>	7,000	19,000
Total	39,200	101,600

Source: Court of Auditors, based on ANR data

3 - Skills and jobs of the future

The “Skills and Jobs of the Future” programme is part of France 2030, which was launched at the end of 2021 with the aim of accelerating training for “jobs of the future” in general (€2 billion for all “jobs of the future” combined). The amount allocated to AI represents 10% of the total budget (€200 million planned, €89 million committed, €44 million disbursed as of 30 June 2025). Four of the projects selected under this programme are being led by 3IA institutes as part of the “French School of Artificial Intelligence”, which aims to develop training supply and certify training courses under a common banner. The goal is to train 120,000

people and raise awareness among 245,000 people by 2028, without it being possible to distinguish between training courses for AI specialists (the focus of the vast majority of projects) and training courses in other disciplines that incorporate the effects of AI on the economic sector concerned.

4 - Other measures

The PEPR IA scheme (€73 million programmed, €62 million committed, €25 million disbursed as at 30 June 2025) aims to accelerate AI research through nine projects bringing together more than 50 French teams, involving researchers and engineers, with a view to promoting an interdisciplinary and collaborative approach. The PEPR AI projects cover several themes: frugal and embedded AI, trusted and distributed AI, and the mathematical foundations of AI. This programme, designed at the end of 2021, reflected the priorities of the time and targeted several communities: addressing trust issues to control the risk of runaway behavior (particularly in terms of software); exploring the niche at the intersection of frugal AI and embedded AI, areas in which the CEA and French industry have strengths. Despite extensive scientific consultation, generative AI did not appear to be a priority at the time.

The “AI Cluster Chairs” scheme (AI Faculty France, €40 million planned), which has not yet been launched, has been postponed to the third phase of the SNIA (AI component of the “Choose France for Science” plan announced in April 2025).

The P16 programme (€8 million planned and committed, €3 million disbursed as of 30 June 2025), strongly supported by INRIA, aims to develop and promote a set of open-source software libraries, in particular Scikit-learn, which has become the most widely used machine learning library in the world. The ambition of P16 is to promote this asset by developing, in particular through the startup Problabl, new modes of transfer to industry within a framework that remains sovereign. INRIA aims to replicate this transfer tool in other areas where France has open-source assets, particularly robotics.

The “maturation-pre-maturation” programme (€7 million planned and committed, half of which was disbursed by 30 June 2025) is based on the principle that the path between research and industrial development needs to be accelerated, whereas other schemes assume the opposite, that the traditional stages of maturation based on a linear view of innovation are poorly suited to AI.

Beyond the measures included in this section, other initiatives presented in the SNIA’s generative, embedded, sober and trustworthy AI sections also contribute to AI research and training, in particular: the extension of the Jean Zay public supercomputer and its network of support engineers, which benefits a wide variety of research projects carried out by public laboratories; the “Digital Commons for Generative AI” initiative, under which several funded projects involve academic partners and national research institutes; the Alliance for Language Technologies (ALT-EDIC), which has a strong research and development (R&D) component, notably through the creation of a unified portal of training databases available to the research community; and, through indirect funding, the National Institute for the Evaluation and Security of Artificial Intelligence, dedicated to the evaluation and security of AI models, which also has a strong R&D component.

B - France's position in AI research and higher education in AI continues to progress, even if several limitations remain

The “AI research and higher education” component of the second phase of the SNIA has made initial progress but has suffered from certain limitations. The late start of most of the initiatives means that it is too early to assess their lasting impact. However, in line with the actions undertaken since 2018, this second phase of the SNIA has undeniably contributed to structuring the French artificial intelligence ecosystem, with a knock-on effect on research and the development of a dynamic network between public, private and academic players.

1 - Initial successes

Several achievements have been made in the “AI research and higher education” component of the second phase of the SNIA.

(a) France's position in AI research and higher education continues to improve. The country has made significant progress in the *Global AI Index*, rising from thirteenth place in the ranking published in September 2024 to fifth place in September 2025¹⁴. In terms of research, training and infrastructure, driven by public actors, France rose from fifth place worldwide in 2021 to third place in 2023 (*Global AI Vibrancy Ranking* published in November 2024¹⁵). France is also the third country in the world in terms of the number of researchers specializing in AI, with more than 4,000 researchers.

(b) With the creation of AI Clusters, the structuring of higher education and research in AI appears more structural in several respects: in accordance with the recommendation made by the Court of Auditors in its April 2023 report, this new certification has made it possible to unify the status of centers of excellence in AI; the central role of universities in the system has been reinforced; AI Clusters also benefit from more precise (more directed) thematic application targeting, which tends to give a local flavor or even a specialization to each “ecosystem” built around these AI Clusters and, subject to effective management, allows for complementarity between the areas of excellence of the different clusters; the integration, in this phase of the strategy and unlike the previous one, of the “higher education in AI” component also tends to reinforce its structural nature, which will need to be assessed over time; the training and research chairs subject to industrial partnerships are aligned with the long-term objectives of the SNIA (frugal, trustworthy, generative AI); strategic partnerships with other European centers of excellence are also beginning to take shape.

(c) Private co-financing of AI Clusters has exceeded expectations. Project financing plans reflect industrial support that is generally well above the minimum threshold of 20% set by the call for projects, demonstrating that the formula meets a real demand.

(d) With regard to the PEPR IA, a scientific tool designed to promote high-level scientific output on selected priorities, initial results show that since its launch it has mobilized 150 researchers and enabled the recruitment of 150 new doctoral students. It also shows that nearly

¹⁴ [The Global AI Index.](#)

¹⁵ [Which countries are leading in AI? | Stanford HAI.](#)

90 publications have been accepted at conferences, two-thirds of which were at leading conferences.

2 - Limitations

However, several limitations remain.

(a) The complexity of France 2030 governance and the expectation of political confirmation at annual events such as Vivatech have resulted, for the main schemes, in excessively long delays between initial announcements and concrete funding commitments, ranging from up to three years in the case of AI Clusters or nearly two years in the case of the PEPR IA. These delays can be explained in part by the government's desire to ensure rigorous evaluation of projects, with several rounds of back-and-forth between project leaders and reviewing departments to guarantee their quality, strategic consistency and structural impact on the ecosystem. During this transitional phase, a continuity mechanism has been put in place to provide bridge financing to enable the 3IA institutes to continue their activities and avoid any disruption to their momentum. This complexity and these delays explain the low rate of budget consumption.

(b) The downward renegotiation of budgets disrupted selection processes, reduced the ambitions and scope of projects, and limited their time frame. In particular, AI Clusters saw their funding reduced by 20% and their duration shortened by two years. In its April 2023 report, the Court already pointed out that the four-year funding model for 3IA centers was too short-term compared to their German equivalents, for example.

(c) The structural nature of AI Clusters remains limited by the fact that funding mainly covers temporary and part-time contract jobs, the permanence of which is not guaranteed. The competitive call for projects approach is not suited to the creation and long-term existence of higher education programmes, as it does not allow for the recruitment of high-level teachers and teacher-researchers to deliver these programmes. The system therefore does not address the issue of permanent staff in a context where AI is becoming a major discipline. For the universities concerned, making the necessary redeployments of permanent posts requires critical mass and long-term management capacity, consisting of setting thematic priorities and avoiding short-term approaches that focus on chasing opportunistic funding. This transition is complex, particularly in a context of institutional autonomy. Incentive mechanisms have been put in place to support this structural transformation, such as making funding for AI Cluster chairs conditional on the availability of qualified permanent staff in AI. Furthermore, recruitment pressures remain high: the private sector, which is highly competitive for AI profiles, offers remuneration levels that pose a real challenge to the attractiveness of academic careers. This structural imbalance calls for a coordinated response at the national level, at the crossroads of research, higher education and innovation policies.

(d) The "Skills and Jobs of the Future" programme, which has a large budget, presents a high risk of fragmentation and a lack of structural transformation. This programme includes funding for skills needs assessments to identify gaps between the skills available and those required in changing sectors, including in the field of AI. However, the initial results of these assessments point to a shared observation: training needs are massive and widespread throughout the economic fabric. This situation argues less for a refined targeting of sectors or

territories than for a strategy of massification of training efforts, in order to enable a growing number of workers, in both initial and continuing training, to acquire the skills necessary for the integration of AI into organizations.

(e) The distribution of roles and the flow of information between the SNIA research pilots (CNIA and INRIA) and the many other actors involved at the national level remains a challenge. This is particularly true for the National Research Agency in its role as operator on behalf of the government, especially in monitoring project performance, beyond the material management of funding and selection processes. The role that INRIA will effectively play in monitoring and evaluating AI Clusters illustrates this challenge. These questions, which concern the governance of France 2030, go beyond the scope of the SNIA alone.

III - France has achieved initial successes in generative AI and in the areas of frugality and trust

Mobilization around generative AI and continued efforts on computing infrastructure have led to visible headway (A). Progress on the issues of frugality and trust is also significant, while results in embedded AI remain mixed (B).

A - Mobilization on generative AI and continued efforts on computing infrastructure have led to visible headway

1 - Overview

In early 2023, at the time of the “ChatGPT revolution”, the development of a French ecosystem for generative AI was identified as a new priority for the SNIA. National coordination responded quickly by adjusting the strategy’s trajectory to include a structural focus on generative AI. Several measures were thus put in place within a tight timeframe. This was particularly the case with the call for projects entitled “Digital Commons for Generative AI”, which aims to strengthen French capabilities in this critical area. Although not specific to generative AI, other SNIA initiatives have contributed significantly to its development, notably the expansion of the Jean Zay supercomputer and the renewal of its network of engineers. €109 million has been committed to this component of the second phase of the SNIA. The rate of consumption of funds was only 46% at the end of the first half of 2025, and only 15% without the Jean Zay supercomputer extension operation (Table 6).

Table n° 6 : Generative AI and computing infrastructure component of the second phase of the SNIA

<i>Measures</i>	<i>Content</i>	<i>Start-up (initial announceme nt of the scheme / first funding commitments)</i>	<i>France 2030 planned budget (in € million and as a percentage of the total)</i>	<i>Commitment s of appropriatio ns as at 30 June 2025 (in € million)</i>	<i>Payments as at 30 June 2025 (in € million)</i>
Digital commons for generative AI <i>Operator Bpifrance, steering DGE, DGRI</i>	Accelerating the creation of digital commons for generative AI	June 2023 / June 2024	58.0 43%	58.0	9
Extension of the Jean Zay supercomputer <i>Operator ANR via GENCI (CNRS, CEA, INRIA), steering DGRI</i>	Increase of the capabilities of the Jean Zay supercomputer	June 2023 / December 2023	40.0 29%	40.0	39.6
Acceleration in the use of generative AI <i>Operator Bpifrance, steering DGE</i>	Supporting partnerships between suppliers and users of generative technologies	April 2024 / Coming soon	27.0 20%	0	0
PNRIA network of engineers <i>Operator ANR, steering DGRI and CNRS</i>	Extension and reinforcement of the Jean Zay supercomputer support engineering team	June 2023 / December 2024	10.0 7%	10.0	1.3
ALT-EDIC <i>Operator ANR, steering Ministry of Culture, DGE</i>	European centre of expertise in engineering and experimentation in generative AI in languages	February 2024 / March 2025	1.2 1%	1.2	0.3
Total			136.2	109.2	50.2
<i>Committed / Planned</i>				80%	
<i>Consumed / Committed</i>					46%

Source: Court of Auditors, based on SGPI and CNIA

2 - The successful shift towards generative AI

In a particularly competitive environment, the national AI coordinator met with key players in the sector, often before they had even set up their businesses and decided to develop them in France. A dynamic interaction began to take shape between researchers specializing in generative AI, innovative companies and investors.

The call for projects “Digital Commons for Generative AI” (€58 million planned and committed, €9 million disbursed as of 30 June 2025) contributed to this structuring, with the establishment of a common dynamic and a catalytic effect, on the one hand, on the financing of French players who were beginning to position themselves in generative AI and, on the other hand, on strategic decisions to develop open foundation models, particularly in the field of imaging.

In just a few months, France has made progress in terms of competitiveness and attractiveness in generative AI, with the emergence of a dozen players covering a variety of specialties, both in terms of tools (text, image and video generation, voice, transformation of textual instructions into concrete actions and deployment of agents), purposes (adaptation of models to needs, dissemination in companies, evaluation and optimization of systems) and sectors (health, defense and security, mobility, education, engineering, biology and chemistry, materials, etc.). France is one of seven countries to have developed AI systems ranked among the top 200 models in the world. Several companies have gained worldwide renown and succeeded in significantly increasing their capital. Decisions to set up operations in France by several Big Tech companies were also taken during the period.

Overall, several rankings agree that France is in the top spot in Europe, including the United Kingdom, in terms of generative AI, reflecting, on a continental scale and without seeking to compare itself to the United States or China, the real success of the research and entrepreneurial ecosystem in this sector.

In addition, the ALT-EDIC alliance, established in February 2024 on France’s initiative, brings together 25 EU Member States. It benefits from total funding of €88 million at European level. Based at the *Cité internationale de la langue française* in Villers-Cotterêts, it has enabled a dialogue to begin between cultural data owners and developers responsible for training generative AI models in compliance with copyright laws.

However, the €27 million “Accelerating the use of generative AI” programme, which is more downstream, has not been implemented.

3 - Accelerating investment in computing infrastructure

In terms of computing infrastructure for training AI systems, mainly generative AI, the capacity of the Jean Zay supercomputer has more than tripled (from 37 to 126 petaflops/s). The investment (€40 million programmed, committed and disbursed under the SNIA, with an additional €5 million from GENCI) was made without interrupting the machine’s operation and while respecting the site’s capacity constraints, including in terms of energy consumption.

Following on from the first phase of the SNIA, the operation was accompanied by financial support from the CNRS, supplemented by the SNIA (€10 million programmed and

committed, €1.3 million disbursed as at 30 June 2025) in order to sustain and strengthen the team of engineers providing support to users. However, securing long-term funding for this support took a long time to materialize, which caused difficulties in managing the staff concerned.

The heat recovered now provides enough energy to heat the equivalent of 1,500 homes (6.5 GWh per year) on the Saclay plateau. The supercomputer's electricity consumption after expansion is estimated at 24 GWh for 2025, representing a budget of around €3.4 million borne solely by the CNRS. The machine's efficiency, expressed as the ratio of its computing power to its electricity consumption (taking into account recycled heat), has thus been multiplied by almost three.

The expanded capabilities of the Jean Zay supercomputer, which recently came into service, are in very high use, with demand estimated at more than five times the available supply. This situation illustrates the rapid acceleration in computing power requirements, particularly for training and evaluating large language models, which mobilize unprecedented volumes of resources. To meet these needs, a new large-scale infrastructure is in the pipeline: the Alice Recoque exascale supercomputer, whose construction by the end of 2026 is co-financed by the cloud acceleration strategy and EuroHPC. The AI Factory France project, led by a consortium of academic players, innovation partners and industrialists, was also selected in March 2025 as the winner of the European EuroHPC programme for the financing of infrastructure support services and their users, including training and economic development that meet the needs of strategic sectors such as health, aeronautics, energy, defense and mobility. The launch of the EU's AI giga-factory programme aims to accelerate this momentum. In addition, a classified strategic military supercomputer for the French army, Asgard, which is completely disconnected from the internet and exclusively dedicated to military use, was inaugurated in September 2025. It is considered to be the most powerful supercomputer of its kind in Europe and the third most powerful in the world, at a cost of €600 million.

4 - The role of the CNIL in ensuring compliance with personal data protection by AI systems

The French Data Protection Authority (CNIL) has invested heavily in the field of AI. In 2023, driven by the combined effects of the rise of generative AI and the introduction of specific European regulations, it adopted an action plan to ensure that AI systems are developed in compliance with personal data protection regulations¹⁶. Numerous monitoring, educational, advisory and practical information initiatives, as well as tailored support for high-potential projects, have been undertaken in this context. As an extension of this action plan, the CNIL has confirmed the central place it accords to AI in its 2025-2028 strategic plan¹⁷.

The CNIL has made significant efforts to engage as early as possible with the research, innovation and business ecosystem and to keep pace with the rapid evolution of innovations and needs. In particular, the practical AI fact sheets it publishes and its recommendations on

¹⁶ Artificial intelligence: the CNIL's action plan | CNIL.

¹⁷ CNIL strategic plan - 2025-2028.

the application of the General Data Protection Regulation of April 2016 (GDPR) to the development of AI systems, with a view to reconciling innovation and respect for individual rights within the legal framework and in a pragmatic manner, are widely used by professionals and the general public alike¹⁸.

The European Data Protection Board (EDPB), which brings together the CNIL and the competent authorities of the EU Member States, adopted an opinion in December 2024 on the processing of personal data for the development and deployment of AI models¹⁹. This opinion expresses an unprecedented harmonized European position on three key aspects: AI models trained using personal data cannot, in principle, be considered anonymous (a case-by-case assessment must be carried out to verify that it is highly unlikely that the individuals whose data was used to create the model can be directly or indirectly identified and that personal data can be obtained from the model through queries); the consent of the individuals concerned prior to the use of personal data for the development and deployment of AI models is not systematically required (legitimate interest may constitute a valid legal basis and several criteria have been identified, in particular whether or not the personal data used is publicly accessible); when an AI model has been developed using personal data that has been processed unlawfully, the lawfulness of its deployment and use is questionable, unless the model has been effectively anonymized. The decision taken in 2025 by a large American company to train its models on European personal data in compliance with the GDPR confirmed that the flexibility allowed by this framework could be compatible with the development of such AI systems.

However, AI companies remain somewhat wary, particularly with regard to the European Regulation on Artificial Intelligence (AI Act), which has yet to be implemented and which adds to existing cross-cutting (such as the GDPR) or sectoral (particularly banking) regulations and introduces a new function for monitoring AI markets. The European Commission and Member States' desire to simplify this framework aims to address this concern. Negotiations over the coming months on the texts implementing the AI Act will be a decisive step, with the challenge of building a comprehensive vision that integrates privacy protection, innovation and operational constraints, and is capable of adapting to the rapid pace of technological change.

Furthermore, in national law, while the GDPR introduced a logic of accountability in 2018 that aimed to eliminate all requests for prior authorization for data processing, "authorization" mechanisms have been maintained, particularly with regard to files used by the police, defense and judicial services, as well as those used for research, studies and assessments in the field of health.

¹⁸ [AI practical information sheets | CNIL](#).

¹⁹ [Opinion 28/2024 on certain data protection aspects related to the processing of personal data in the context of AI models | European Data Protection Board](#).

B - Progress on the issues of frugality and trust is real, but the results in terms of embedded AI remain mixed

1 - Overview

In addition to generative AI, the second phase of the SNIA planned to place particular emphasis on three priorities involving specific disruptive technological innovations (deep tech): embedded, frugal and trustworthy AI. These different aspects of the strategy respond to industrial policy objectives in sectors where France already has strengths: embedded AI is a requirement of the transport industry, with a potential comparative advantage over the United States and China; the objectives of frugality and trust, in addition to being in line with European societal choices, contribute to the competitiveness of AI models and their dissemination. A total of €185 million has been committed to these components. The consumption rate of the corresponding appropriations stood at 52% at the end of the first half of 2025:

Table n° 7 : Embedded, frugal and trustworthy AI component of the second phase of the SNIA

<i>Measures</i>	<i>Content</i>	<i>Start-up (initial announcement of the scheme / first funding commitments)</i>	<i>France 2030 planned budget (in € million and as a percentage of the total)</i>	<i>Commitments as at 30 June 2025 (in € million)</i>	<i>Payments as at 30 June 2025 (in € million)</i>
Call for embedded AI projects <i>Operator Bpifrance, steering DGE</i>	Development of AI components for embedded technologies	November 2021 / May 2003	65.6 30%	65.6	18.3
IPCEI microelectronics and connectivity <i>Coordination DGE (electronics strategy)</i>	European collaboration for the development of advanced chips	2022 / July 2023	50.0 23%	50	50
Assessment and safety issues and security of models <i>Operator not yet designated, steering SGDSN and DGE</i>	Creation of INESIA (INRIA, LNE, ANSSI, PReN)	January 2025 / forthcoming	36.0 16%	0	0
TEF Edge AI <i>Operator ANR, steering DGE</i>	Funding for research infrastructure for decentralised computing (PREVAIL winning consortium)	November 2021 / December 2022	24.9 11%	24.9	8.7
DIAT <i>Operator Caisse des dépôts, steering MTE Ecolab</i>	Demonstration of frugal AI solutions for ecological transition of local authorities	June 2023 / July 2023	18.7 8%	18.7	3.3
DeepGreen <i>Operator Bpifrance, steering CEA, DGE</i>	Open source platform for embedded and trusted AI	April 2023 / November 2023	18.4 8%	18.4	9.2
Sectoral TEF <i>Operator Bpifrance, steering DGE and the Ministries of Agriculture and Health</i>	Financing of four TEF (Transition Energy Fund)	November 2021 / July 2023	7.4 3%	7.4	5.8
Total			221.0	185.0	95.3
<i>Committed / Planned</i>				84%	
<i>Consumed / Committed</i>					52%

Source: Court of Auditors, based on SGPI and CNIA – SGDSN: General Secretariat for Defense and National Security and MTE: Ministry for Ecological Transition

2 - Embedded AI

The situation remains fragile with regard to embedded AI. Initiatives taken by the SNIA have included a call for projects entitled “Technological maturation and demonstration of embedded AI technologies”, the selection of 12 projects as part of the Important Project of Common European Interest (IPCEI) on electronics and connectivity, the European PREVAIL project to develop new chip prototypes, and the DeepGreen open-source platform for embedded and trusted AI, the real impact of which remains to be seen.

Several French hardware manufacturers are recognized as being among the best in Europe, but this sector remains under pressure, particularly with regard to the supply of hardware and components, where companies are struggling to stand out from international competition and raise funding. On the demand side, however, the adoption of embedded AI solutions is progressing rapidly, particularly among large companies.

3 - Frugal AI

Significant progress has been made in promoting the emergence of less energy-intensive AI models. This includes the development of small language models, the creation of a general reference framework for frugal AI by the French Standardization Association (AFNOR), which is set to become an international standard, and the commitment of a dozen local authorities to the development and deployment of energy- and data-efficient AI systems to support their ecological transition, as part of the “Frugal AI Demonstrators for Ecological Transition in Local Areas” (*Démonstrateurs d’IA frugale au service de la transition écologique dans les territoires*, Diat) initiative. More broadly, measuring environmental impact is a systematic requirement in the specifications for SNIA programmes. It is also worth recalling that the Jean Zay supercomputer is one of the most environmentally responsible in the world, thanks to its internal cooling system and its mechanism for recovering waste heat to heat the equivalent of 1,500 homes on the Saclay plateau, a first in Europe on this scale.

At the AI Action Summit held in Paris in February 2025, a coalition for sustainable AI was launched²⁰. Initiated by France, in collaboration with the United Nations Environment Programme and the International Telecommunication Union, it brings together around 100 participants, including companies, research institutes, non-governmental organizations and associations, investors and public bodies, with the support of some 20 countries and international organizations. The coalition is proposing an ambitious research programme involving French stakeholders. A hackathon combining AI and energy efficiency, the Frugal AI Challenge, was also organized during the summit.

4 - Trustworthy AI

When it comes to trustworthy AI, France is at the forefront of several European and international initiatives. It has won all five calls for tenders co-financed by the EU concerning “AI Testing and Experimentation Facilities” (TEFs), infrastructures that enable developers to test (free of charge for startups) their models and verify their regulatory compliance in the areas

²⁰ Coalition for Sustainable AI.

of health (TEF-Health), agriculture and food (AgrifoodTEF), smart territories (CitCom.AI) and agile production (AI-MATTERS).

France is coordinating the only project aimed at establishing “Union Testing Facilities” to assess compliance with the AI Act (TestAI project). It is also the only EU country that is a member of the International Network of AI Safety Institutes (which follows on from the Bletchley Park Summit in November 2023 and the Seoul Declaration for Safe, Innovative and Inclusive AI in May 2024) and has set up an AI Safety Institute, the National Institute for the Evaluation and Safety of Artificial Intelligence (*Institut national pour l'évaluation et la sécurité de l'intelligence artificielle*, INESIA). This institute, co-managed by the General Secretariat for Defence and National Security (*secrétariat général de la défense et de la sécurité nationale*, SGDSN) and the DGE, brings together, without creating a new legal structure, the national players involved in AI evaluation and safety (ANSSI, INRIA, the National Metrology and Testing Laboratory – *Laboratoire national de métrologie et d'essais*, LNE – and PEReN) with the mission of promoting security, inclusiveness and trust in AI.

INRIA has also led the development of a leaderboard for comparing the performance of large language models adapted to the French language and culture²¹.

The launch by the Government in May 2024 of work on the trusted digital solutions sector is in line with these initiatives, with the aim of deploying sovereign and competitive integrated data-storage-AI (Cloud-Data-AI) offerings, streamlining the digital solutions market to make it more open to European companies, and developing and disseminating training modules common to the trusted digital sector. Similarly, the cloud strategy, which identifies cloud technologies as essential for ensuring digital sovereignty and remaining at the forefront of innovation, aims in particular to develop a “Trusted Cloud Services” label.

The “defense and security” dimension of trusted AI is also essential in order to effectively combat harmful uses of AI, not so much by targeted sectors (the European directives list eleven highly critical ones) as by the criminal use that can be made of it: disinformation, fraud, cyberattacks, and the manufacture of new nuclear, radiological, biological or chemical weapons using AI. Within the SGDSN, the Viginum service, responsible for vigilance and protection against foreign digital interference, uses AI to analyze inauthentic online behavior and information operating methods that could harm France’s fundamental interests. A partnership between Viginum and INRIA was concluded in March 2025 on these issues.

International initiatives, in which France has played a leading role, have also contributed to the promotion of frugal and trustworthy AI.

IV - France has begun to play an active role in AI on the European and international stages

European policy on artificial intelligence accelerated significantly in 2025, with France playing a key role (A). More broadly, France’s leading role on the international stage was confirmed by the success of the AI Action Summit held in Paris in February 2025 (B).

²¹ https://huggingface.co/spaces/fr-gouv-coordination-ia/llm_leaderboard_fr#/ (The Future of AI).

A - European policy on AI accelerated significantly in 2025, with France playing a key role

Since 2018, the EU has gradually developed a wide range of instruments aimed at promoting and regulating the development of AI. Regulatory aspects have been the focus of attention, with the entry into force of four pieces of legislation that are unparalleled in the rest of the world: the GDPR in 2018, the Digital Market Act (DMA) and Digital Services Act (DSA) in 2023, and the AI Act in 2024. The Commission's initiatives have also aimed, particularly from 2024 onwards, to support data and computing infrastructures, startups and innovation, as well as research and training. These priorities are similar to those of the SNIA, and France has not been immune to this acceleration. In terms of research, the CNRS played a leading role in the adoption in July 2025 of a joint declaration by the six main European research organizations, calling for an ambitious European strategy for AI in the service of science²².

In an August 2024 report, the European Court of Auditors assessed the effectiveness of the Commission's contribution to the development of the European AI ecosystem²³. It made substantial criticisms and recommendations that contributed to the reorientation of European policy in favor of AI in 2025. In particular, the Court noted a lack of coordination between the Commission and Member States, insufficient monitoring of the performance of investments and actions to support research and innovation, and weak private co-financing. There was also a growing awareness of the need to simplify the regulatory framework, which risked hampering innovation and the spread of AI within the EU compared to other less regulated competitive markets.

European AI policy accelerated significantly in 2025. In February, the Commission announced the InvestAI initiative, which aims to mobilize €200 billion in investment in AI within the EU, including through a new €20 billion European fund for AI gigafactories. In April, it then presented the outlines of its new programme, the AI Continent Action Plan²⁴. This plan is based on two interrelated pillars: competitiveness and productivity (with the idea that AI-based automation, optimization and decision-making support will drive productivity gains and innovative business models, thereby strengthening Europe's economic prosperity) and sovereignty, security and democracy (technologies play an increasing role in the global balance of power, and AI is essential to Europe's security and the safeguarding of democracy). This very broad plan is structured around five strategic areas: IT infrastructure, data, skills, AI adoption and regulatory simplification.

This action plan was supplemented in October 2025 by the European Commission's presentation of two strategies: firstly, the Apply AI Strategy, which aims to increase the adoption of artificial intelligence, including by SMEs, to strengthen the competitiveness of strategic sectors and the technological sovereignty of the EU²⁵ and, secondly, the Strategy for

²² <https://www.cnrs.fr/en/update/artificial-intelligence-g6s-recommendations>

²³ European Court of Auditors, *Special Report 08/2024: The EU's challenge of artificial intelligence – No progress possible without stronger governance and more and better targeted investment*, May 2024 [Special Report 08/2024: The EU's challenge of artificial intelligence](#).

²⁴ [Action Plan for the AI Continent | Building Europe's Digital Future](#).

²⁵ [Implementing the AI Strategy | Building Europe's Digital Future](#)

AI in Science, which aims to place Europe at the forefront of research and scientific excellence based on artificial intelligence²⁶.

However, the concrete implementation of these various announcements remains subject to the adoption of the next European multiannual financial framework.

B - France's leading role on the international stage was confirmed by the success of the Paris AI Action Summit in February 2025

1 - Bilateral cooperation

At the bilateral level, cooperation with Germany is a priority, but not exclusive. It began to take shape in the research field during the first phase of the SNIA. The Ministry of Higher Education and Research, together with its German counterparts, has identified three priorities for possible enhanced bilateral cooperation in the future, relating to attractiveness, mobility and talent training, computing, and the link between research and innovation. A Franco-German dialogue between AI industries was organized in January 2025 with a view to supporting European technological sovereignty, and the two countries shared a joint stand on AI (French-German Tech Lab) at the Vivatech trade fair in June 2025. Several measures dedicated to AI are also included in the Franco-German Economic Action Programme adopted at the end of August 2025 by the Joint Council of Ministers of the two countries²⁷.

2 - Multilateral cooperation and the February 2025 summit on AI

A collaboration between the OECD and the Global Partnership on AI was initiated by France in 2023 and has become one of the main forums for international AI governance. In 2024, the three GPAI centers of expertise (France/INRIA, Canada, Japan) also signed a tripartite agreement establishing a joint working platform.

French diplomacy has also been very active in the work initiated by the United Nations on AI, which notably led to the conclusion in September 2024 of a Global Digital Compact annexed to the Pact for the Future, which calls in particular for strengthening the international governance of AI²⁸.

Co-chaired by France and India, the AI Action Summit, held on 10 and 11 February 2025 in Paris, brought together representatives from more than 100 countries. Preceded by two days of scientific discussions, the summit was organized around five themes: AI for the public good, the future of work, innovation and culture, trustworthy AI, and global AI governance. A declaration was adopted by some 60 countries²⁹, the EU and the African Union Commission, which the United States and the United Kingdom did not sign. Many other texts were adopted, including the Charter on AI for the Public Good³⁰, the New Partnership for Promoting AI for

²⁶ [European Strategy for Artificial Intelligence \(AI\) in Science | Research and innovation](#)

²⁷ [Franco-German economic action programme. | Élysée.](#)

²⁸ [n2425290.pdf.](#)

²⁹ [Declaration on inclusive and sustainable artificial intelligence for people and the planet. | Élysée.](#)

³⁰ [Paris Charter for Artificial Intelligence in the Public Interest. | Élysée.](#)

the Public Good³¹, the Coalition for Sustainable AI³², the Commitment to Trustworthy AI in the World of Work³³, the Declaration on Maintaining Human Control in AI-Enabled Weapons Systems³⁴, and the Co-Chairs' Statement on International Governance of AI³⁵. Many other actions were initiated³⁶, including the creation of a new international foundation on AI in the public interest, Current AI, to promote issues related to data, openness and accountability. The creation of an observatory on the energy impact of AI, under the auspices of the International Energy Agency (IEA), was also announced.

Finally, the summit provided an opportunity to promote French excellence in AI. The launch of a third phase of the SNIA was announced following the meeting of the interministerial committee on AI that preceded the summit³⁷, as well as an acceleration of private investment in AI in France (€109 billion announced for the coming years, mainly for the financing of data and computing centers for AI).

V - The challenge of massification and support the spread of AI beyond the circle of specialists has not received sufficient attention

However, too little progress has been made in several key areas to date. The challenge of massification and support for the dissemination of artificial intelligence beyond the circle of specialists – businesses, public administrations, students, citizens – has not received sufficient attention. The priority of supporting demand for AI solutions from businesses has only benefited from very modest measures (A). The delay in adapting all initial and continuing training programmes to AI has not been made up (B). The transformation of public action through AI, which was not a priority in the second phase of the SNIA, has remained very modest (C). Actions targeting regions and a broad audience have also not been a priority for the SNIA to date (D).

A - The priority of supporting demand for AI solutions from businesses has only benefited from very modest measures

All of the actions in the second phase of the SNIA aimed to accelerate the spread of AI throughout the economy by training talent, developing sovereign technology tailored to the needs of businesses, and providing shared platforms and open-source building blocks that act as catalysts for the adoption and integration of AI solutions. However, the dissemination of a general technology tends to occur gradually, often over several decades, with stages of invention, innovation, adoption and then dissemination. It requires redoubled efforts on all

³¹ [New partnership to promote AI for the public good | Élysée.](#)

³² [Coalition for Environmentally Sustainable Artificial Intelligence. | Élysée.](#)

³³ [Pledge for a Trustworthy AI in the World of Work | Élysée.](#)

³⁴ [Paris Declaration on Maintaining Human Control in AI-Enabled Weapon Systems. | Élysée.](#)

³⁵ [Co-Chairs' Statement on International Artificial Intelligence Governance. | Élysée.](#)

³⁶ [ce43652059af11d9563431de957f7e9d5fbc888a.pdf.](#)

³⁷ [Making France an AI powerhouse | info.gouv.fr.](#)

complementary levers – skills, infrastructure, support, uses – to ensure that the economic and social benefits of this transformation are fully realized across the productive fabric.

The issue of supporting the adoption of AI solutions by businesses must be distinguished from that of developing the AI economic sector. On the first issue, there is still a huge amount of work to be done. On the second, France has made significant progress: more than 1,000 French startups are active in the field of AI in 2025, twice as many as in 2021; these companies raised nearly €2 billion in 2024; 16 French unicorns (startups valued at over \$1 billion) incorporate AI into their value proposition; several major French groups are expanding their AI offerings and increasing their investment in AI research. France ranks first in Europe in terms of the number of foreign investment projects in AI. It is also the leading European host for research and decision-making centers of AI Big Tech companies.

1 - The AI Booster scheme

A single instrument has been set up under the SNIA to promote the adoption of AI solutions by businesses, with limited resources: the AI Booster scheme, which aims to support individual businesses in their first steps towards artificial intelligence (€10 million planned and committed, €7.5 million spent for the benefit of final beneficiaries as at 30 June 2025).

This demand-driven support scheme is an exception to the supply-driven France 2030 doctrine. An initial budget of €25 million was earmarked to partially cover the cost of services provided by Bpifrance to businesses, but this was reduced to €10 million in 2023 when the SNIA budget was revised. The underlying decision was to direct resources towards schemes likely to maximize the overall impact on the spread of AI in the economy. Investments in French AI startups were therefore prioritized, as their potential for rapid and widespread deployment was considered to be more beneficial for the economy as a whole.

Implemented by Bpifrance, AI Booster is a support programme for French SMEs and mid-cap companies wishing to integrate AI solutions to increase their productivity and enrich their offerings. The roll-out of this programme in 2024 showed that it takes 8 to 10 months to move from intention to a concrete project ready for implementation. Three levels of support are offered: raising awareness, acculturation and training for managers of SMEs and mid-cap companies on the benefits of AI (since the launch of the programme, more than 20,000 people have participated in webinars and online training courses, and nearly 1,000 companies have benefited from the online self-assessment); fee-based individual data and AI diagnostics service, aimed at highlighting priority areas for value creation through AI and identifying use cases (around 700 SMEs and mid-cap companies have initiated this diagnostics process, which generally lasts two to three months and has enabled 100% of cases to identify simple and inexpensive use cases); fee-based individual service aimed at qualifying AI solutions that meet identified needs and establishing an implementation plan (around 120 companies that completed the previous phase used this service to support structural projects). A fourth level of support was envisaged but was not implemented due to a lack of resources: this was a consulting service for testing the AI solution, aimed at initiating its operational deployment.

The companies receiving support come from a variety of sectors and have adopted a wide range of AI solutions. The programme's gradual and personalized approach meets the expectations of entrepreneurs who, without the support of this scheme, would not have

embarked on such a structured and rapid reflection on the integration of AI into their businesses with a view to improving their operational performance and competitiveness, enriching their offering and modernizing their operating methods.

The AI Booster scheme initially generated considerable interest. However, the National Coordination for AI considers that its impact remains limited and its results mixed, insofar as the approach adopted does not allow for scaling up and the scheme struggles to address a genuine market failure. Its budget has not been replenished.

2 - Other initiatives

Several complementary measures aimed at promoting the adoption of AI by businesses have been implemented under the impetus of the National Coordination for AI, although they are not funded by the SNIA. Led by the DGE, these initiatives have made it possible to develop a set of tools to facilitate the integration of AI solutions by businesses in response to their operational and sectoral needs. For example, in early 2025, the “AI for Efficiency” call for expressions of interest enabled nearly 600 respondents to share their feedback on the adoption of AI, and an “AI Observatory” was set up to share a map of use cases solved by AI, with the idea that the dissemination of this information would, through natural market competition, trigger an effective engine for adoption.

However, international comparisons³⁸ show that French companies are not particularly advanced in this area, and the various initiatives undertaken are still in their infancy and appear modest in relation to the challenge, even though, due in particular to informal channels of diffusion, France is now one of the five countries in the world where the use of AI is most widespread overall³⁹.

B - The delay in adapting all initial and continuing training programmes has not been made up

Alongside specialized higher education in AI, the challenge of deploying AI across all training programmes – higher education in all fields, continuing education, school education – appears to be twofold: AI calls for a transformation not only of teaching approaches and tools, but also of the content of teaching, including to take into account the transformation by AI of the professions for which these training programmes prepare students.

The first phase of the SNIA did not address training issues, and the second phase did not prioritize the adaptation of all initial and continuing training programmes to AI. The recommendation made by the Court on this point in its April 2023 report has not been implemented.

In terms of higher education outside the specialized AI sector, a few multidisciplinary initiatives have begun to be taken within the framework of the AI Clusters and the CEI “Skills and Jobs of the Future”. However, the number of multidisciplinary initiatives supported by the

³⁸ See, for example, *2023-2024 figures for the global AI market - Bpifrance Le Hub*.

³⁹ Microsoft AI Economy Institute, *AI Diffusion Report: where AI is most used, developed and built*, November 2025.

latter scheme, which is designed on a bottom-up basis, remains marginal. The very small proportion of sectoral training courses adapted to AI (known as "AI+X" or "X+AI" schemes, depending on the place of AI training in the course concerned) is not closely monitored. INRIA has also taken the initiative, through its "Learning Lab", to produce a number of MOOCs⁴⁰ dedicated to AI.

The challenges of continuing education have not been adequately addressed either. The challenge remains immense, both for general continuing education in AI and for the acquisition of the more specific skills required by the transformation of a given sector, profession or function through AI. The project for a platform to consolidate continuing education in digital technology (French Digital Skills Hub), led by INRIA, has not been developed due to a lack of funding.

The issue of adapting school education to AI has also not been included in the scope of the SNIA's France 2030 funding. Outside the SNIA, the Ministry of National Education's digital education department has launched a number of programmes to promote the dissemination of AI, some of which date back to the predictive AI stage, and has worked on defining a framework for the use of AI in education, published in June 2025. There are also a number of scattered experiments, most often the result of individual initiatives, but which are generally neither structured nor commensurate with the scale of the challenges. Awareness is growing and demand for AI training is becoming more widespread.

In its May 2025 report on AI in schools⁴¹, the General Inspectorate of Education, Sport and Research emphasized that pupils' rapid appropriation of the many uses of AI is challenging assessment practices and pedagogy, with AI *"arousing both fears and hopes among teachers"* in terms of personalization and differentiation of learning. The mission considers that school is *"the place where AI training can and must be developed, enabling all pupils and future citizens to make critical, ethical and reasoned use of it"*. In particular, it recommends the rapid implementation of *"a massive training plan for executives"* and *"an AI training curriculum that will enable both the training of future specialists and the education of all students on the challenges and uses of AI."* It also considers it *"imperative to develop a sustainable economic model to enable scaling up."*

A mission on AI in higher education teaching practices also published its findings in July 2025⁴². The mission notes in particular that the use of AI is developing in higher education institutions in a heterogeneous manner and mainly at the individual level, in both teaching and administrative management, in the absence of a shared framework for use and in a context of lack of technical capacity, with risks of inequality or lack of personal data protection. It notes that awareness-raising and X+AI training, which are essential for adapting the use of AI to the context of each discipline, are very underdeveloped. Around thirty recommendations are made based on six objectives: training trainers and students; adopting AI in higher education institutions and in society; transforming universities in the age of AI; developing infrastructure and technical solutions; pooling content and best practices; and promoting a national policy for the adoption of AI in education. The mission estimates that the funding required to implement

⁴⁰ *Massive Open Online Courses*, i.e. online courses open to all.

⁴¹ [Artificial intelligence in schools | Ministry of National Education, Higher Education and Research.](#)

⁴² [Education and artificial intelligence: two reports submitted to ministers | Ministry of National Education, Higher Education and Research.](#)

its recommendations will amount to between €300 and €500 million for the first five years. This fiscal envelope has not yet been allocated.

In June 2025, the Ministry and the Agency for the Pooling of Universities and Institutions (*Agence de mutualisation des universités et établissements*, AMUE) launched an innovation partnership with a major French AI company to develop an AI conversational agent to meet the specific needs of students, teachers and administrative staff in higher education.

As in higher education, the dissemination of AI across all areas of research, particularly scientific research, remains a challenge, beyond specific AI research. The “AI for Science and Science for AI” (AISSAI) center, an internal tool of the CNRS, was created for this purpose in 2021. In many fields, the ability of researchers to introduce high-performance AI solutions tailored to their specific needs will be critical to the future of their work. Several Nobel Prizes and eminent international awards have thus begun to recognize scientific work achieved with the help of advanced AI systems.

No strategic thinking has been undertaken in the area of continuing education. The Ministry recognizes that continuing education in AI is struggling to develop within higher education institutions and that reflection is needed in order to develop a sustainable model that is both competitive and complementary to what is offered by the private sector.

C - The transformation of public action through AI, which was not a priority in the second phase of the SNIA, has remained very modest

1 - Modest actions taken

Unlike the first phase of the SNIA, the second phase did not identify the transformation of the public sector through AI as one of its priorities. The theme is not included in the France 2030 doctrine, nor has it been prioritized in fiscal decisions on other envelopes. Furthermore, from 2023 onwards, no funding has been allocated to this issue, even though it had been identified as essential as early as 2018.

Interministerial initiatives led by DINUM, ministerial or local initiatives have been taken, but they have been small in scale, scattered and with limited effects in terms of transforming public policies and services.

In an October 2024 report, the Court of Auditors examined the case of the Ministry of Economy and Finance, where no fewer than 35 AI programmes had been deployed since 2015 to detect fraud risks among individuals and difficulties faced by businesses, or to provide faster responses to users. The Court recommended strengthening the management of these programmes to ensure trustworthy public AI, better assessing the productivity gains generated and allocating them in a transparent manner, and better anticipating and sharing the consequences of these AI programmes on the tasks entrusted to civil servants⁴³.

⁴³ Court of Auditors, *Artificial intelligence in public policy: the example of the Ministry of Economy and Finance*, final observations, October 2024.

At the interministerial level, in line with the government's digital roadmap of March 2023, the DINUM's AI strategy was based on three key areas: integrating AI functionalities into everyday digital tools; deploying mature off-the-shelf solutions that meet the identified needs of government departments; and developing tailor-made AI solutions. It aimed for an approach that favored short experimentation cycles, the pooling of resources, and the use of free software and open tools, while ensuring compliance with the principles of ethics, security, and sovereignty. In practice, however, in the absence of interministerial priority on public transformation through AI since 2023 and in a context of overall reduction in the state budget, DINUM has sought opportunities for low-cost instruments, pooling and the creation of reusable commons.

Among the AI projects to which DINUM has contributed since 2023 are ALLiaNCE, an incubator that aims to structure public investment in AI, and Albert API, an infrastructure that allows administrations to access open-source AI models to automate synthesis, classification, document search and transcription (it is used in more than 200 public digital products and receives 10,000 requests per day; however, there is a lack of objective data to assess its actual impact), and Compar:IA, a tool for comparing conversational language models in French (the platform, which opened to the public in October 2024, receives over a thousand visits per day and is now used by major model publishers).

2 - Other limitations

Beyond the lack of structural funding and the modest achievements over the 2023-2025 period, the assessment of AI deployment in public administrations reveals several limitations.

(a) The government has made very little use of public procurement to promote the adoption of AI in the public sphere. In particular, DINUM has favored internal developments, with the risk that the solutions proposed may not always be as effective as those available on the market, particularly from French software publishers. Work is underway on the issue of developing public procurement in AI, and an interministerial call for expressions of interest was launched in 2025 to assess the relevance of AI solutions available on the market for the needs of public services.

(b) The preference for free software cannot be systematic nor take precedence over the search for sustainable solutions and the use of innovative public procurement to strengthen the market, particularly in France, for technology suppliers. An ecosystem policy, involving all public and, above all, private stakeholders in a balanced manner, has yet to be developed. The search for open data sources may also appear limiting, as it is not always suited to the needs of administrations.

(c) Several projects developed by DINUM did not produce satisfactory results, could not be scaled up, or were not designed to be shared. Albert France Services, the first AI experiment conducted by DINUM in 2023, which aimed to test a generative AI conversational assistant to help France Services advisers, revealed critical limitations in terms of the quality of responses and the traceability of sources. Several products, developed in particular within the framework of LabIA and which had demonstrated their relevance during the experimental phase, struggled to achieve national or interministerial deployment. Several projects funded in recent years by

Etalab⁴⁴ have provided relevant solutions for targeted use cases, but without building truly shareable building blocks.

(d) The rollout of INRIA's Apollo programme has not been a priority. This programme for transfer to the public sector, at the crossroads of public policy demands and sovereign AI provision, is part of an innovative public procurement approach and would be likely to improve the quality of public policies and generate significant savings. It is included in the roadmap of the new strategic sector committee (*comité stratégique de filière*, CSF) for "Trusted Software and Digital Solutions".

D - Actions aimed at regions and a wide audience have not been a priority for the SNIA to date

The SNIA has not prioritized action aimed at regions or structuring its relationship with local authorities.

More and more local authorities, particularly at the regional level, are tending to develop their own strategies for artificial intelligence. Several regions have launched roadmaps to support innovation in this field, as well as training and dissemination of AI in their territories. However, there is no formal framework for coordination between the SNIA and the regions on AI, nor is there any coordination between the regions themselves.

Nor has the SNIA provided an opportunity to define a roadmap for communicating with the general public about the economic and societal challenges associated with AI. Statements on the subject of AI at the highest level of government have so far focused more on France's excellence in AI research, innovation and startups than on the economic and societal upheavals that this revolution is bringing about. Events such as the Paris AI Action Summit in February 2025 received significant media coverage, but this remained sporadic.

⁴⁴ Department of the Interministerial Directorate for Digital Affairs, Etalab coordinates the design and implementation of the government's strategy in the field of data.

CONCLUSION

Launched without prior evaluation, the second phase of the SNIA was supposed to address the challenge of mass adoption and support for the diffusion of artificial intelligence in all areas. Announced in November 2021, this phase has seen its priorities, budget and timetable change significantly due to growing constraints on public finances and the need to reallocate resources to support the development of generative AI, an issue that had not been anticipated on the eve of the “ChatGPT revolution”.

In total, the government allocated €1.1 billion over the period 2023-2025, which is one-third less than initially announced, and the slow start-up of most of the measures has resulted in a low level of budget consumption (35% as of 30 June 2025). The governance of public policy on artificial intelligence has also remained complex, despite some positive developments and several examples of successful coordination with other so-called "acceleration" strategies of the France 2030 programme.

The initial results of this second phase are beginning to emerge in several areas. Although it is still too early to fully assess the effects, the initiatives taken to strengthen the structure and excellence of research and higher education in the field of AI are producing initial results, and France’s position in this area is improving. Our country has thus risen from thirteenth place in the Global AI Index published in September 2024 to fifth place in September 2025. In terms of research and training in artificial intelligence, France ranks third in the world. More than 4,000 French researchers are currently working on AI.

The mobilization that this second phase has enabled in the field of generative AI has also borne fruit. At the beginning of 2023, France had only one player positioned in this type of system. In just a few months, French industry has made progress in terms of competitiveness and attractiveness, with the emergence of a dozen players operating in a wide variety of fields. The number of French AI startups has doubled since 2021: more than 1,000 of them are active in this field in 2025 and they raised nearly €2 billion in funds in 2024. Sixteen French unicorns incorporate artificial intelligence into their value proposition and several large French groups are increasing their offering and investment in AI research. France is the leading European country in terms of the number of foreign investment projects in artificial intelligence, and in terms of hosting research and decision-making centers for global AI leaders.

Efforts to develop computing infrastructure have continued, with the expansion of installed capacity and investment in a new-generation supercomputer. Significant progress has also been made on the issues of frugality and trust. Finally, France is no stranger to the acceleration of European policy on AI and, more broadly, to the fact that international forums are addressing key issues of governance and regulation of the development of artificial intelligence. The success of the AI Action Summit held in Paris in February 2025 confirmed France’s special place on the international stage.

However, alongside these successes, several equally important areas have been neglected. The challenge of massifying and supporting the dissemination of artificial intelligence beyond the circle of specialists – businesses, public administrations, students, citizens – has so far received too little attention, even though it was at the heart of the ambitions set out for this phase of the SNIA and the years 2023-2025 were critical in this regard. As a result, the priority of supporting business demand for AI solutions has only benefited from very

modest measures, and the expected acceleration and massification of the diffusion of artificial intelligence in the economy has not taken place. The delay in adapting all initial and continuing training programmes to AI has not been made up either, even though this is an area where the stakes are considerable and the risks high. Essential projects concerning schools and universities have yet to be designed and implemented. The transformation of public action through artificial intelligence, which has also not been a priority, has remained very disappointing too: despite isolated initiatives, the administration is generally lagging behind. Finally, actions targeting local areas and the general public have not been a priority for the SNIA to date, even though they appear all the more necessary as the impacts of this general-purpose technology accelerate, intensify and become more widespread.

Chapter III

The outlook: consolidating the successes of public policy on AI and broadening its scope

When launching a third phase of the SNIA, several prerequisites must be met (I). The successes of the first two phases of the SNIA must be built upon to achieve a more structural transformation (II). Five critical challenges, which have not been sufficiently addressed to date, must be placed at the heart of public policy on artificial intelligence (III).

I - When launching a third phase of the SNIA, several prerequisites must be met

A third phase of the SNIA was launched in February 2025 (A). Successfully scaling up to meet the challenges of the AI revolution requires several prerequisites to be met (B).

A - A third phase of the SNIA was launched in February 2025

A third phase of the SNIA was announced at the end of the February 2025 interministerial AI committee meeting, focusing on four priorities: strengthening computing infrastructure and critical links in the AI value chain; training and attracting AI talent to cultivate the best competitive advantage; accelerating the use of AI; and providing the means to build trustworthy AI. Each objective was accompanied by the announcement of a series of measures supported by France 2030⁴⁵. Several announcements followed, on the sidelines of the AI summit in February 2025, then at the Choose France and Vivatech trade shows in May and June, and finally with the “Dare to Use AI” plan in July, each time providing an opportunity to refine the strategy and clarify the outlines of the main measures of this third phase.

The “Dare to Use AI” (“Osez l’IA”) plan aims to accelerate the spread of AI in businesses⁴⁶. Based on the observation that “AI is a lever for competitiveness, with an expected gain of 20% in productivity per company” and that “only 13% of French SMEs and mid-cap

⁴⁵ See the [press kit released after the meeting of the Interministerial Committee on AI on 6 February 2025, Making France an AI powerhouse.](#)

⁴⁶ *Dare to Use AI: a plan to spread AI throughout all businesses* | Ministry of the Economy, Finance and Industrial and Digital Sovereignty.

companies” have adopted AI solutions, the plan sets the ambition of making AI an “*accessible, concrete and useful tool for all French companies, wherever they are, regardless of their sector, by 2030*” (with the objectives that 100% of large French groups, 80% of SMEs and mid-cap companies, and 50% of very small businesses use AI by 2030).

The plan is structured around three areas. Area 1, “*raising awareness to promote the adoption of AI*”, includes the creation and mobilization of a network of 300 “AI Ambassadors”, in conjunction with the Francenum portal, chambers of commerce and industry, Bpifrance, French Tech and various networks across the country. Axis 2, “*training through an AI academy*”, provides for the launch, by the end of 2025, of a platform offering training courses and tutorials tailored to each audience and the implementation of training courses at all levels, in particular through the “Skills and Jobs of the Future” scheme, with the aim of training 15 million professionals by 2030. Finally, priority 3, “*supporting businesses in identifying relevant AI solutions*” includes an AI data diagnostics scheme, managed by Bpifrance, to enable SMEs and mid-cap companies to identify and exploit the potential of AI within their businesses, with 40% of the costs covered (in line with the AI Booster scheme), a catalogue of solutions and use cases, state-guaranteed loans (via Bpifrance) for SMEs undertaking structuring AI projects, and the “AI Pioneers” scheme to support the most ambitious and transformative projects likely to lead to technological breakthroughs in key sectors.

The “AI Pioneers” scheme (€100 million planned) is a particular innovation. Jointly supported by INRIA and Bpifrance, its aim is to promote the emergence of AI applications in strategic sectors where the solutions offered to date have not yet led to real leaps in terms of productivity, scientific innovation or transformation of uses. The underlying idea is to finance disruptive public-private projects, removing technological barriers, with the potential to transform industrial sectors, such as the AlphaFold software, the result of a partnership between a major private company and two public laboratories, which has led to a major breakthrough for the chemical and pharmaceutical industry, earning its developers the Nobel Prize in Chemistry in 2024.

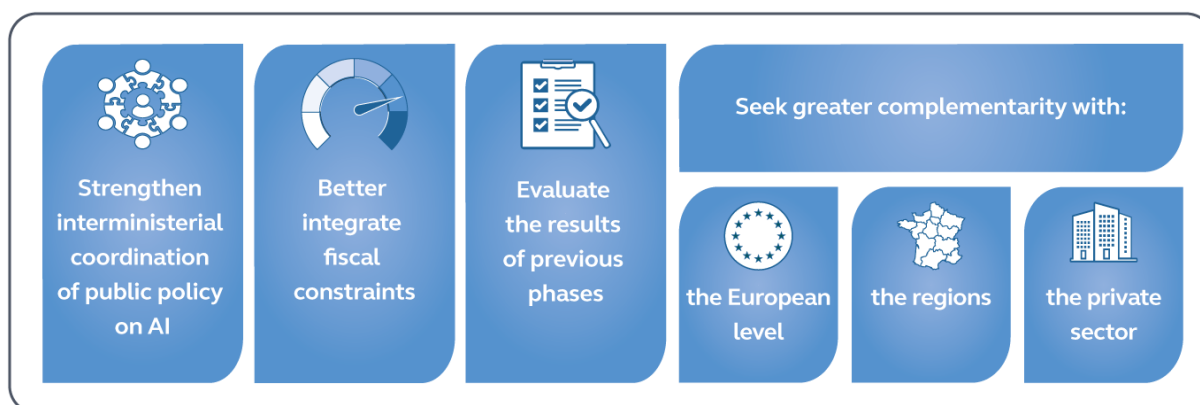
At the same time, the following measures were announced: the transformation of the National Digital Council into the National Council for AI and Digital Technology (*conseil national du numérique en conseil national de l’intelligence artificielle et du numérique*, Cn), an independent collegial body responsible for informing public decisions; the strengthening of cloud offerings, with support for French suppliers to develop an offering tailored to AI (€100 million, already provided for in the cloud acceleration strategy); the acceleration of investment in public and private computing capacities in line with European initiatives (€70 million, also already planned in the cloud strategy), with in particular the development of Europe’s largest AI campus (1.4 GW capacity) by a joint venture between Bpifrance, an Emirati investment fund and two major AI players; the AI Faculty France initiative, part of the Choose France for Science programme for June 2025, to fund research and teaching chairs in AI conditional on the creation of permanent positions (€40 million, a measure carried over from phase 2 of the SNIA); and support for disruptive innovation in chips and middleware (€25 million).

B - Successfully scaling up to meet the challenges of the AI revolution requires several prerequisites to be met

AI is no longer the same issue it was in 2018, when the national strategy was launched. The stakes have changed: the revolution brought about by this general-purpose technology is

accelerating, intensifying and amplifying, reaching a magnitude with which few other technological breakthroughs in history can be compared. AI is no longer a matter for specialists alone; it affects all fields of knowledge, the economy, and society, and is becoming an essential priority for public policy. Successfully achieving the scale change required by the AI revolution requires several prerequisites to be met:

Schéma n° 1 : Prerequisites for successfully scaling up AI



Source: Court of Auditors

1 - Strengthen interministerial steering of public policy on AI

The performance of AI, in terms of both the quality of its results and the variety of domains and functions it covers, continues to accelerate. The challenges posed by this acceleration are also becoming increasingly complex. The place of AI in 2025 will be vastly different from what it was in 2018, or even in 2022. The conclusions of the April 2025 Artificial Intelligence Index, report by Stanford University (United States), provide an overview of the spectacular advances made in recent months⁴⁷.

Given the scale of the AI revolution, the *status quo* is not an option: it is no longer possible to consider that public policy on AI can be treated in isolation, as one of many acceleration strategies of the France 2030 programme, or even as a simple component of the government's digital strategy. The challenge of AI, which has dominated the digital agenda for the past two years, also extends far beyond it: it impacts most public policies. Awareness and a change in approach are needed.

In this context, it seems essential to strengthen interministerial steering of public policy on AI⁴⁸. To this end, it is essential to:

- favor a broad strategic approach, integrating all digital issues (covered by the scope of the digital programmes agency led by INRIA) as well as “upstream” (components) and “enabling” aspects (energy, connection networks, data, regulatory aspects, etc.), in all their

⁴⁷ *Stanford Institute for Human-Centered Artificial Intelligence (HAI), Stanford University, Artificial Intelligence Index Report 2025, April 2025.*

⁴⁸ See recommendation no. 1.

dimensions (training, research, innovation, economy, regulation, transformation of public policies, etc.);

- significantly strengthen the direction of public policy on AI by replacing the current national coordination with a General Secretariat for Artificial Intelligence (*secrétariat général à l'intelligence artificielle*, SGIA) reporting directly to the Prime Minister and responsible for all public policy on AI, across a broad scope, and for the associated interministerial budget. This SGIA should have authority over the various entities and operators involved in this policy, lead a community made up of all stakeholders, in particular by steering a network of ministerial AI coordinators, and be equipped, through redeployment, with a genuine interministerial team bringing together a variety of skills capable of covering the entire field of public policy on AI;
- organize the comitology of public policy on AI around the future SGIA so that it can derive maximum benefit from it. In particular, the newly established National Council for AI and Digital Technology should be placed under the SGIA; its composition should also be enriched and structured into several panels so that it reflects all the dimensions that AI public policy must address⁴⁹;
- clarify the role of the entities responsible, under the authority of the SGIA, for implementing the sectoral aspects of public policy on AI (the digital programmes agency run by INRIA for training, research and innovation, the DGE for businesses, the DINUM and the DITP for the modernization of public administrations) and the operational deployment of the instruments of this policy (in particular prefects, universities, national research operators and Bpifrance)⁵⁰ ;
- prioritize the implementation of public AI policy through pilot projects rather than through fragmented calls for projects.

⁴⁹ The following should be established:

- a “users” panel (representing users of AI systems: businesses, public administrations, civil society), with the aim of feeding into the “demand” side of public policy on AI (tasks to be automated as a priority, barriers to be removed to promote the spread of AI in the economy, training to be put in place, etc.);
- a “developers” panel (representing suppliers of AI models, components, data centres, supercomputers, investors, etc.), with the aim of informing the “supply” side of public policy on AI (providing information on major trends and innovations, identifying digital commons to be put in place, detecting vulnerabilities in supply and sovereignty across the entire AI sector, etc.);
- an “evaluators” panel (composed of entities involved in AI control and evaluation activities: regulatory authorities, accreditation, certification and labelling bodies, testing centers, etc.), with the aim of informing the ‘governance’ aspect of public policy on AI;
- an “anticipation” panel (representing research and higher education organizations, professional sectors, social partners, economists, sociologists and humanities and social sciences theorists, etc.), with the aim of anticipating the changes brought about by AI in society and guiding technological choices in the service of the common good.

⁵⁰ The ANR, for its part, could see its role gradually refocused on non-directed generic calls for projects to bring new topics to the fore in a bottom-up approach (beyond the SNIA, the ANR already implements and monitors non-directed calls for projects and, since 2018, has funded 720 projects dealing with AI for a total of €340 million).

2 - Better integrate the challenges of fiscal sustainability and the effectiveness of AI policy

As in all areas of public action, the government's policy on artificial intelligence must be compatible with the trajectory of public finances.

The government has not been able to meet its fiscal targets for the first two phases of the SNIA, with an average of around €250 million committed each year over the period 2018-2022, compared to the €310 million initially planned, and €280 million over the period 2023-2025, compared to the €520 million initially planned. Announcements concerning the third phase appear to fall well short of these amounts, even though the stakes appear higher than ever. In times of fiscal restraint, there may be a strong temptation, for example, to consider that we have “arrived” in terms of AI research and higher education, or to postpone critical investments and “massification” projects that are essential for scaling up. Furthermore, in certain areas, a change of approach seems inevitable: government alone will not be able to finance the intensive investments required by the AI revolution, particularly in terms of data centers and computing infrastructure.

Strategic thinking must be undertaken from this perspective, under the authority of the future SGIA, in order to better integrate the challenges of fiscal sustainability and AI policy effectiveness⁵¹. While ensuring that significant public investment capacity remains available, the challenge is to define guidelines that enable public AI policy to contribute more effectively to fiscal recovery, in particular by aiming for greater efficiency of instruments and greater subsidiarity between actors, and by promoting use cases that enable greater efficiency in public action and combat fraud. This involves:

- seek greater efficiency in the instruments selected: reducing delays to make them more compatible with the pace of AI; streamlining procedures by modifying certain aspects of the France 2030 doctrine;
- prioritize options that reinforce sustainable and inclusive prosperity, quantifying the “return on investment” for public finances in terms of avoided social expenditure and expected additional fiscal revenue;
- develop new forms of partnership with the private sector, not only for investment in data and computing infrastructure (the movement initiated at the February 2025 summit must be continued), but also in their operation (which requires the definition of an appropriate economic model);
- maximize the leverage effect of every euro of public money invested;
- seek greater subsidiarity within the public sphere by better coordinating European and national strategies, giving greater responsibility to state operators and relying more on local authorities, particularly regions, in their areas of competence;
- among the priorities for public transformation through AI, promote use cases that enable greater efficiency and combat tax and social security fraud in a targeted and regulated manner, quantifying their impact in terms of restoring public finances;

⁵¹ See recommendation no. 1.

- invest more in public policy levers other than fiscal ones, particularly legal levers, to influence the definition of international standards and European and national regulations on AI.

3 - Evaluate the results of previous phases of the SNIA and draw inspiration from best practices abroad

Before embarking on the third phase of the SNIA, the government did not carry out an in-depth evaluation of the first two phases of this strategy, let alone of all the public measures that contribute to public policy on AI. Such an evaluation appears essential in order to assess the impact of the various measures put in place in relation to the objectives pursued.

An AI commission, co-chaired by Anne Bouverot, Chair of the Board of Directors of the École Normale Supérieure, and Philippe Aghion, professor at the Collège de France and economist, was tasked with considering the future of public policy on AI: its report, submitted to the President of the Republic in March 2024, does not claim to be an evaluation, but has made several observations and recommendations, that this report will return to⁵².

Nor was an in-depth comparative analysis of AI policies implemented in other countries carried out to inform the choices for the third phase of the national strategy. No recent benchmarking of this nature has been carried out by the OECD, even though the Organization has conducted several studies on AI. In particular, an “OECD Expert Group on the Future of AI” was set up to help policymakers proactively consider and manage the changes brought about by AI⁵³ and a comparative analysis was conducted on the use of AI in a dozen key functions of public administrations, based on 200 use cases⁵⁴. The United States’ July 2025 AI action plan, “Winning the Race”, which is based on three pillars – innovation, infrastructure, and international diplomacy and security – is also worthy of attention⁵⁵.

It seems essential to conduct, in the short term, an in-depth assessment of the results of the main measures that have contributed to public policy on AI since 2018 and a study of best practices in the most competitive countries. This assessment, involving, where appropriate, the relevant interministerial and ministerial inspection bodies and France 2030 operators, will be essential for rationalizing public spending and the governance of this public policy. It must lead to the adoption and dissemination of performance indicators to enable adjustments to be made where necessary. This imperative is reinforced by the need to adapt public action to the rapid developments in these technologies⁵⁶.

⁵² Artificial Intelligence Commission, *AI: Our ambition for France*, March 2025. *A report on seizing the opportunities of artificial intelligence* | info.gouv.fr

⁵³ *OECD, Assessing potential future artificial intelligence risks, benefits and policy imperatives*, November 2024.

⁵⁴ *OECD, Governing with Artificial Intelligence: Current Status and Prospects for Core Government Functions*, September 2025.

⁵⁵ *America's AI Action Plan*.

⁵⁶ See recommendations 1 and 10.

4 - Better coordination with the European level

The scaling up of artificial intelligence policy means making better use of the European level. This is all the more necessary given that the geopolitical context has changed, giving new prominence to the issues of sovereignty and strategic autonomy of the European Union. Europe still lags far behind the United States and China in the AI race: a significant effort is needed to catch up.

The logic of seeking EU co-financing or “rates of return”, which can have the perverse effect of destabilizing a national strategy if it is articulated inconsistently and with a short-term perspective, must be overcome. French and European AI strategies would benefit from being designed and implemented in a more coordinated manner, so that the aspects that make more sense – particularly in terms of critical mass – at the continental level can be managed or coordinated at that level, and conversely, so that the EU is not tempted to cover aspects that appear more relevant – in accordance with the principle of subsidiarity – to be managed at the national or even sub-national level for reasons of proximity.

This reasoning applies to each of the strategic areas of the European Action Plan for AI of April 2025: IT infrastructure, data, skills, algorithm development and AI adoption, and simplification of rules. The strategic coordination around the AI factory programme is a success that should be replicated across the board.

The new SGIA will need to fully integrate this European dimension. The challenge is to establish, by the next AI summit in February 2026, a more structured link between national and European strategies and to deepen the partnership with Germany on AI-related issues⁵⁷, which requires:

- much more structured interaction, as early as possible, between, on the one hand, the authorities steering national AI policy (including, on the higher education and research side, the digital programmes agency, for which strengthening European cooperation is one of the key priorities) and, on the other hand, the authorities responsible for European policy (the General Secretariat for European Affairs (*Secrétariat général aux affaires européennes*, SGAE), France’s Permanent Representation to the EU, the European Commission and in particular its Directorate-General for Communications Networks, Content and Technology, the European Parliament, etc.);
- the definition by France of its priority expectations in terms of European AI policy and a strategy to ensure that they are taken into account;
- the structuring, on the basis of a strategic roadmap, of the bilateral Franco-German partnership on AI issues, which is set to play a leading role within the EU. In particular, cross-investments are to be developed on upstream issues, and the rise of binational players must be encouraged on downstream issues.

⁵⁷ See recommendation no. 1.

5 - Better mobilization of local authorities

In the regions, public initiatives in the field of AI are multiplying in an uncoordinated manner. Until now, the SNIA has not made coordination with local authorities a priority. Some regions are ahead of the curve, particularly those that benefit from centers of excellence such as AI clusters, investments in data centers or computing infrastructure, or a dynamic fabric of innovation and AI companies, while other regions risk being left largely behind.

Structuring appears essential to better mobilize regions, and this structuring must be put in place by the new SGIA. The challenge is twofold: on the one hand, to ensure that public policy on AI benefits everyone; on the other hand, to define a division of roles that allows for coordination and complementarity between international, European, national and local actions, in order to enable real added value from interventions and funding, particularly those deployed at regional level⁵⁸. To this end, several initiatives appear useful:

- organizing an annual “AI conference”, a political consultation forum made up of representatives of the State and local authorities, to discuss the objectives and results of public policy on AI;
- disseminate guidelines to local authorities on the areas of action to be prioritized at each level, whether local initiatives or measures supporting national policy. Structural measures should be prioritized to avoid the risk of a proliferation of micro-counters. The dissemination of good practices to a network of AI correspondents in the regions would also be useful;
- integrate AI public policy priorities into State-Region planning contracts, adapting them to each region (particularly in terms of the local higher education and research ecosystem and the industrial fabric) to enable complementarity between actions and public funding in the regions;
- More generally, in steering national AI policy, structure the relationship with prefects and heads of regions. This must be done by fully involving all stakeholders in the regions, particularly, in the area of higher education and research, universities and regional centers of INRIA.
- Develop AI-related initiatives in the regions (such as frugal AI demonstrators supporting the ecological transition in the regions) and better integrate the regional dimension into the design of all AI public policy instruments.

6 - Redefine the division of roles with the private sector

The respective roles of the public and private sectors in the development of AI are atypical and tend to evolve rapidly as technology advances: private research, particularly within the large American firms that dominate the digital market, has played an essential role in the development of AI, both in terms of innovation (with very short lead times between laboratories and startups) and in terms of funding volumes; data centers, particularly clouds, which are essential for AI systems, are overwhelmingly owned by private companies and associated with many equally essential services (AI model training, education, etc.); computing infrastructures,

⁵⁸ See recommendation no. 1.

which are growing in size, will also be increasingly supported by private investors; the components used for AI are the exclusive domain of a few private manufacturers; AI software solutions, including those that dominate the market and drive future R&D priorities, are also largely private; the largest digital companies and a few private investment funds are also responsible for almost all of the financing for the growth of AI startups.

Before setting priorities for public policy on AI, it seems essential to seek to take advantage of these dynamics. A strategic review must be undertaken at the level of the new SGIA to redefine the relationship with the private sector in achieving objectives of general interest on AI-related issues, in order to better leverage the potential of private provision, identify the most appropriate forms of state intervention and maximize the leverage effect of each public euro invested⁵⁹. This involves, in particular:

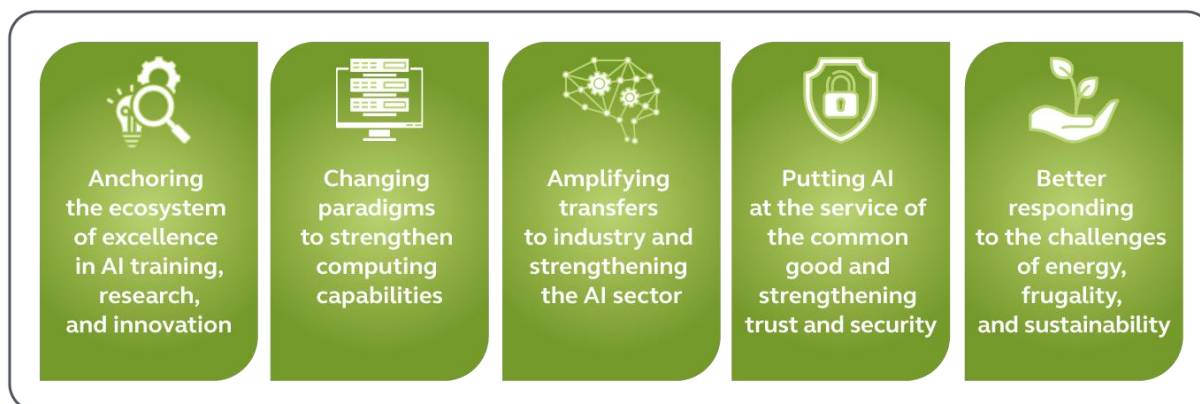
- defining the best approaches to achieving objectives of general interest in the field of AI, distinguishing between: (i) cases where private supply, possibly encouraged and supervised by the State, appears sufficient (e.g. areas of innovation in AI that appear to be sufficiently covered by private research); (ii) new types of public-private partnerships or a transition to private ownership need to be developed (e.g. for the operation of future AI mega-factories); (iii) the use of public procurement involving sovereign actors could be developed (e.g. to accelerate public transformation through AI); (iv) issues of sensitivity or sovereignty require specific regulation of private supply or correction of market mechanisms (e.g. for the management of certain data, or for the development of upstream sectors for AI components); or (v) private sector intervention appears to need to be limited (e.g. on certain issues related to the governance of trustworthy AI) ;
- in each case, examine the most appropriate form of state intervention: regulation (e.g. on certain issues related to the frugality of AI systems), incentives (e.g. for the adoption of AI by businesses), contractual relationships with private partners (e.g. for the operation of future AI mega-factories), direct intervention (e.g. for the massification of training);
- systematically consider how to maximize the leverage effect of public intervention (a particularly critical issue, for example, in relation to the adoption of AI by businesses).

II - The successes of the first two phases of the SNIA must be built upon to achieve a more structural transformation

It would be a mistake to consider that the priorities where the SNIA has achieved initial success no longer require attention. AI remains a rapidly developing field, and public policy in this area must not “let its guard down” but, on the contrary, seek to amplify its areas of excellence and aim for more structural transformations and impacts. Five key areas are concerned by this deepening:

⁵⁹ See recommendation no. 1.

Schéma n° 2 : Areas for further development for more structural transformations



Source: Court of Auditors

A - Anchoring the ecosystem of excellence for training, research and innovation in AI

One of the SNIA's key assets has been the creation and structuring of an ecosystem of excellence in AI training, research and innovation. Public policy on AI must continue to focus its efforts on this priority. AI cannot be developed as an industrial sector, but rather in a way that allows very different players to develop fluid interactions facilitated by the state. It calls for a responsive approach and a developed appetite for risk, bringing together public officials, centers of excellence in training and research, AI startups, tech companies, large innovative groups, venture capitalists, investment banks, etc.

The steering of public policy on AI, under the direction of the new SGIA, in close collaboration with INRIA and Bpifrance in particular, must preserve this multidisciplinary approach. A new step must be taken in this area to enable a more structural transformation. This requires several levers⁶⁰:

- conducting a forward-looking exercise to define quantitative and qualitative objectives for permanent staffing needs in public AI research and higher education ("forward-looking management of jobs and skills in higher education and research in the digital sector") and, more broadly, in scientific professions (engineers, mathematicians, computer scientists, etc.). These objectives will need to be implemented at the level of universities, national research institutes and engineering schools, and integrated into recruitment decisions in order to enable the structural anchoring and massification of centers of excellence in AI, in particular AI Clusters. INRIA's digital programmes agency, in conjunction with universities, is called upon to play a key role in the implementation of this component;
- adopting an investment approach to AI research, which gives universities and national research organizations a multi-year vision, ensuring consistency over time and enabling the transition to sustainable models and funding. This involves: moving away from the mechanism of calls for projects; incorporating policies of excellence in AI research and higher education, and their link with innovation and businesses, into the objectives,

⁶⁰ See recommendation no. 2.

resources and performance contracts of universities and national research institutions; conducting regular performance reviews (on-the-spot and ex post). The challenge is to move away from short-term thinking (press releases, opportunistic rush for funding, etc.) to long-term accountability for stakeholders, in order to capitalize on structured projects to support startups, create partnerships with businesses (drawing inspiration from the governance of the cybersecurity transfer programme), and increase the visibility of French excellence in AI within the EU and internationally.

- Include research programmes, particularly scientific research programmes, in the objectives of public policy on artificial intelligence, as has already been done for AI clusters and the PEPR IA. One ambition must be to better anticipate technological breakthroughs such as agentic AI, i.e. the ability of an AI system to achieve a specific objective with limited supervision, and to strengthen interdisciplinarity between AI and priority areas such as engineering, health, energy, law, the environment and finance. This means giving the government back a say in the purpose of public research in AI. Dual research, such as that begun by INRIA with the Ministry of the Armed Forces, must be developed;
- gradually refocus the work of the ANR on non-directed mechanisms (generic calls for projects, an essential complement to public research policy);
- encourage private or partnership-based research in AI, in particular open research on objectives of public interest, including by promoting two-way mobility and reinforcing private research as a strategic investment for businesses;
- strengthen bridges between research and business more broadly, while guaranteeing academic independence and freedom;
- conduct a mission on the “attractiveness of AI talent”, in order to enhance the attractiveness and working conditions of AI researchers (a more systemic measure of a general nature may also be desirable).

B - Changing the paradigm to strengthen computing capabilities

The first two phases of the SNIA made strengthening computing capacities for AI a high priority, and the successes achieved in this area (extension of the Jean Zay supercomputer and support programme by a team of engineers) are set to see further developments from 2026 onwards (creation of an exascale supercomputer, Alice Recoque, as part of the EuroHPC initiative, and the AI Factory France project).

However, demand for computing infrastructure for training, fine-tuning and production (inference) of AI systems is growing at a much faster pace than supply, calling for an acceleration of public and private investment. Significant private investment in France was announced at the February 2025 summit and detailed in the months that followed.

The next step planned as part of the European strategy, with a budget of €20 billion (InvestAI fund), is to build “AI Gigafactories” equipped with around 100,000 latest-generation chips and four times more powerful than “AI Factories”, including the Alice Recoque supercomputer. In this context, France aims to increase its public high-performance computing capacity to 1.2 exaflops/s, multiplying its current power sixfold and thus joining the leading pack in terms of equipment for training next-generation AI models.

This change of scale calls for reflection, under the leadership of the new SGIA, to better coordinate the needs of European players in terms of access to computing, but also to redefine the role of the state, which will no longer be able to finance the bulk of these infrastructures, and more generally to rethink the positioning of the public computing offer for AI. A paradigm shift is needed. France and, more generally, the EU will have to move from a period where the main challenge was to open up public infrastructure to private users to a period where the conditions of access for public and private users other than the owners of the infrastructure to private computers, which are also partly supported by public funds, will have to be defined. This will require the development of new forms of large-scale public-private partnerships⁶¹.

C - Amplifying transfers to industry and strengthening the AI sector

A change of scale appears necessary with regard to: (i) instruments to promote the transition from research to industrial development (“from the laboratory to the startup”), (ii) the structuring of relations with AI companies, and (iii) policies to support the growth of AI companies, including startups involved in disruptive technological innovation⁶².

With regard to the first aspect (transfer of AI research to industry), massification involves, in particular:

- the ramp-up and expansion of support mechanisms for the creation of software-intensive technology companies, such as the INRIA Startup Studio, to reach volumes that are no longer counted in tens but in hundreds of startups. The digital programmes agency could play a decisive role in scaling up such transfer mechanisms;
- the strengthening of the focus of research on industrial innovation, in particular through partnerships between research and French companies, including within AI clusters;
- giving preference to AI partnership research with French companies. Such an objective requires a delicate balance between excellence in AI research and industrial sovereignty strategy;
- continuing to build large open data consortia, such as the P16 / Probabl model.

With regard to the second aspect (structuring the relationship with the AI business sector), a major step forward was taken with the establishment in April 2025 of the CSF “Trusted Software and Digital Solutions”. This is the first recognized structure for the sector that creates AI technologies (€24 billion in turnover in 2023, growing by more than 10% per year). This sector can also make a decisive contribution to sovereignty objectives, compared to large French digital services companies, which mainly sell American solutions. It is essential that national AI policy be aligned with this Trusted Digital CSF.

With regard to the third aspect (growth of AI companies), the lack of depth in the national and European capital markets, particularly venture capital, constitutes a structural limitation compared to the United States that goes beyond AI policy. The September 2024 report on the future of European competitiveness by Mario Draghi, former President of the European Central Bank, calls for an unprecedented investment effort in the AI sector, particularly in ten strategic

⁶¹ See recommendation no. 3.

⁶² See recommendation no. 4.

sectors⁶³. The growth of the AI economic sector in the United States has relied less on the venture capital industry than on investments by digital giants. Europe, meanwhile, is struggling to keep up with the high-volume capitalization rounds of AI companies, although some observers hope that the growth of AI companies will be an opportunity for a change in investment culture. Risk-taking is high, returns on investment are uneven, as is the ratio between the market valuation of certain AI companies and the revenue they generate or expect to generate, raising fears of a bubble that could lead to major adjustments, including in Europe, in the coming months.

In its aforementioned report, the AI commission recommended redirecting part of French people's savings towards innovation, by mobilizing life insurance taxation tools, and creating a €10 billion investment fund for corporate private equity and public support for AI. The government did not adopt these proposals but, as announced in February 2025, this strategic sector for France will be supported by Bpifrance to the same extent, i.e. €10 billion by 2029.

In addition to efforts to promote the growth of AI companies through sovereign equity investments, several complementary initiatives could be pursued, in particular:

- leveraging public procurement by formalizing a government purchasing policy in the field of AI. This lever, which creates conditions enabling French companies to submit competitive bids, appears to be both more efficient than direct aid mechanisms and better suited to meeting the expectations of AI startups. However, compared to other countries, France has made very little use of it to date. Scaling up in this area would also help accelerate the transformation of government agencies. Tailor-made treatment should be reserved for public procurement involving the processing of sensitive data. Finally, France should exert its influence on the ongoing redefinition of the European framework to encourage the emergence of sovereign players;
- strengthen the monitoring of industrial acquisitions in the AI sector, where the growth of startups presents the risk that high investments in the initial phase will be lost at the time of acquisitions enabling a change of scale;
- increase the leverage effect of public investment in AI, including within the Bpifrance portfolio, in particular by targeting the allocation of resources that are not already dominated by American players;
- mobilize institutional investors to acquire stakes in AI companies, for example by strengthening initiatives such as the Tibi 2 fund managed by the French Treasury, and increase influence in initiatives aimed at deepening the capital and venture capital market for AI at European level;
- exploit the industrial levers enabled by the development of dual-use AI products for civil and military purposes, ensuring that the Ministry of the Armed Forces' strategy, within the framework of the military programming law, is aligned with the broader framework of interministerial policy on artificial intelligence.

⁶³ *The Draghi report on EU competitiveness.*

D - Putting AI at the service of the common good and strengthen trust and security

The first two phases of the SNIA, particularly the second, have already seen numerous initiatives launched to promote the development of trustworthy AI, including at international level, and to advance assessment and security issues, which also have implications for the dissemination of AI and industrial policy. However, the French remain far from convinced about artificial intelligence. A recent survey shows that the majority remain wary (64% have a negative opinion, compared with 54% in Germany and 50% in Italy) and that only 17% of those surveyed believe that AI will make their work easier and more interesting, compared with 59% who fear that this technology will threaten their jobs⁶⁴.

Efforts must continue in this area⁶⁵. In particular, it is necessary to:

- implement certain initiatives such as INESIA and the commitments made at the February 2025 AI Action Summit through several declarations or the launch of the Current AI foundation;
- beyond that, define an international strategy for trustworthy AI, so that France's position continues to influence issues of evaluation, security, ethics, and standardization (how, with a view to establishing AI standards, can standardization bodies be influenced and public and private actors brought together?).

More broadly, as identified in the Villani report of March 2018, public policy on AI must be geared towards the common good. The choices made in public policy on AI will largely determine the effects of AI development in terms of sustainable and inclusive growth, employment and the evolution of social inequalities. On these key issues, public policy on AI, under the authority of the new SGIA supported by an expanded National Council for AI and Digital Technology⁶⁶, must define a clear strategic direction. In the spirit of the declaration adopted at the February 2025 summit, it appears necessary to:

- conduct in-depth economic studies on AI in order to take an informed step back and better guide AI technologies and uses in line with the objectives of inclusive and sustainable prosperity;
- upgrade the tools that provide upstream information to public policymakers and organizations on the impacts of AI deployment on the economy and society, in particular the LaborIA laboratory on the world of work.

⁶⁴ Cevipof confidence barometer Wave 16 February 2025

⁶⁵ See recommendation no. 5.

⁶⁶ See recommendation no. 1.

E - Better respond to challenges related to energy needs, frugality and sustainability of AI

The challenges related to the energy requirements, frugality and sustainability of AI constitute a final major area of focus, in which the efforts undertaken in the first two phases of the SNIA, particularly with regard to frugality, must be continued and amplified⁶⁷.

The spectacular expansion of AI is accompanied by major challenges related to its electricity consumption and ecological footprint. Observers particularly criticize the schizophrenia that may exist in the European AI strategy between the promotion of frugality objectives and the launch of AI giga-factory programmes. Several studies highlight the spectacular increase in the energy consumption of AI systems. The IMF estimates in a May 2025 study that, worldwide, data centers already consume the equivalent of France's electricity, and that this figure could triple by 2030⁶⁸. There is a real risk of a global slowdown in AI development in the medium term due to a bottleneck in electricity capacity. The tech giants are well aware of this and are starting to invest in nuclear technologies that could meet their electricity needs.

In the longer term, there is a risk of creating an environmental time bomb. At the February 2025 summit, the IEA announced the creation of an observatory to monitor the impact of AI on energy consumption. Reconciling technological innovation, performance, energy constraints and respect for the environment is a technological challenge, but also a societal one, requiring coordination between researchers, manufacturers and regulators to establish standards and solutions that promote sober and responsible AI.

France has strengths in terms of energy infrastructure and expertise on issues of frugality. By continuing its efforts, it can gain a decisive competitive advantage. To achieve this, under the authority of the new SGIA, it is necessary to:

- assess the energy and technological dependencies that weigh on the future of AI development in France (and Europe), but also our country's competitive advantages in terms of energy, infrastructure and sober and sustainable AI systems;
- better integrate energy and environmental issues into all aspects of public policy on AI and better coordinate them with the European level and, conversely, coordinate AI-related issues in strategies relating to energy and the ecological transition;
- explore different ways of guiding users of AI systems towards solutions that are tailored to their needs and as energy-efficient as possible, and inform them about the energy consumption of their AI uses;
- define an international influence strategy to increase transparency on energy consumption issues in data centers and AI, in cooperation with the IEA, and to promote energy efficiency targets for AI systems.

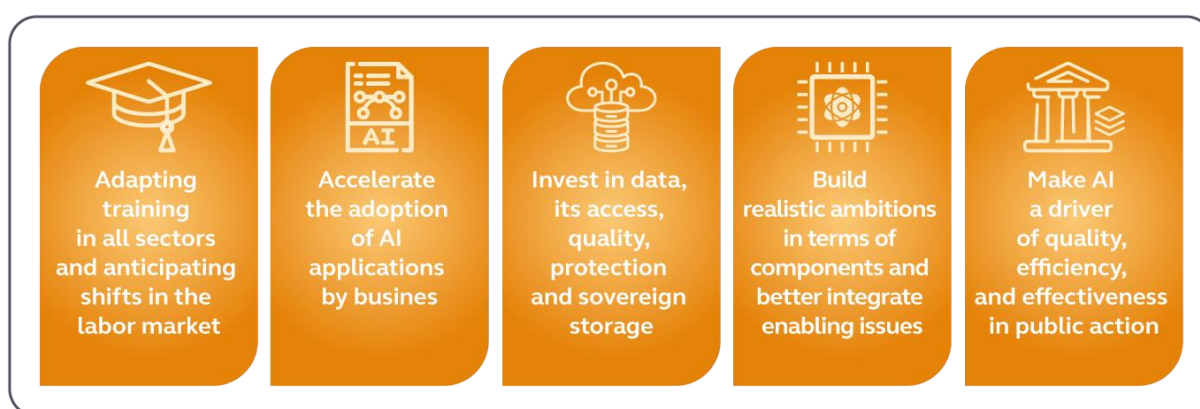
⁶⁷ See recommendation no. 5.

⁶⁸ *AI Needs More Abundant Power Supplies to Keep Driving Economic Growth.*

III - Five critical challenges, which have not been adequately addressed to date, must be placed at the heart of public policy on AI

The first two phases of the SNIA have revealed blind spots in several key areas, or at least have failed to achieve results commensurate with the challenges, particularly in sectors that require the involvement of a wider circle than just AI professionals and experts. Five critical challenges, which have not been sufficiently addressed to date, must be placed at the heart of public policy on AI:

Schéma n° 3 : Critical challenges to be placed at the heart of public policy on AI



Source: Court of Auditors

A - Adapting training in all sectors and anticipate changes in the labor market

AI requires new skills and is profoundly changing most occupations, including the most skilled and high value-added ones. Several recent IMF studies⁶⁹ have highlighted the consequences of AI deployment on the labor market and skills requirements, as well as the changes in training that these changes entail. The IMF points out in particular that, whereas automation and technology have traditionally affected repetitive tasks, AI stands out for its ability to impact highly skilled jobs. It estimates that, in advanced countries, AI could have a major impact on around 60% of jobs (according to this estimate applied to France, this means that around 13 million workers could be affected); for around half of these, the most 'secure' jobs, the integration of AI would lead to productivity gains through complementarity, provided that suitable and competitive AI applications are deployed quickly and employees are trained to use them in the best possible conditions; for the other half, those jobs that are more exposed due to low complementarity, AI applications could perform essential tasks currently performed

⁶⁹ See in particular [Carlo Pizzinelli et al., *Labour Market Exposure to AI: Cross-country Differences and Distributional Implications*, October 2023](#), and [Mauro Cazzaniga et al., *Gen-AI: Artificial Intelligence and the Future of Work*, January 2024](#).

by humans, reducing the demand for labor, lowering wages and reducing hiring or even eliminating jobs.

Several recent studies have highlighted the already significant impact of generative AI on the US labor market, particularly on early-career workers in occupations most exposed to automation by artificial intelligence. An initial study concludes that there has been a relative decline of 13% since the end of 2022 in the employment of young people aged 22 to 25⁷⁰. A second study shows that the emergence of consumer generative AI at the end of 2022 marked a turning point in the labor market and that, after a period of stagnation, employment among young graduates declined from the first quarter of 2023 onwards in companies that had adopted artificial intelligence, with an 8% drop after 18 months⁷¹.

In the case of France, a January 2025 study⁷² estimates that by 2030, due to the acceleration of AI deployment, and in particular generative AI, which now extends to areas requiring creativity and critical judgement, nearly 27% of tasks performed by employees could be entrusted to AI. Depending on how quickly these technologies are adopted by the French economy, the study estimates that the deployment of AI could lead to between 1.2 and 1.7 million job changes by 2030, affecting between 4% and 6% of French employees currently in work, with increased productivity, increased competitiveness and economic growth potential (up to 3% potential annual productivity growth in France by 2030 if AI technologies are adopted quickly and automated working hours are fully redeployed). In French companies that are focusing on retraining, nearly a third of the workforce could be retrained internally, nearly a quarter of the needs could be met by hiring, and nearly a fifth by outsourcing. Anticipating and managing these professional changes and transformations requires a detailed understanding of the skills that will be needed.

Adapting the world of work to these changes requires an unprecedented effort in initial and continuing training, primarily in sectors that are already heavily impacted by AI applications. For example, AI-related transformations in the medical field, in all specialties and at all stages of the care pathway (from diagnosis, including medical imaging, to treatment, patient and disease monitoring, prevention, and the development of new therapies, medical techniques and drugs)⁷³, are leading to significant changes in the skills expected of doctors and require a thorough rethinking of the content of their training.

A second critical issue for higher education and school concerns the impact of AI on teaching, both in terms of deploying AI to modernize teaching tools, which poses many challenges (educational value of the tool, adoption by teachers, data protection, nature of the contract with the supplier, etc.), but also more generally in terms of adapting to the fact that young people are increasingly using AI, particularly widely available generative AI

⁷⁰ Erik Brynjolfsson, Bharat Chandar, Ruyu Chen, Stanford University, *Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence*, August 2025.

⁷¹ Guy Lichtinger, Hosseini Maasoum, Harvard University, *Generative AI as Seniority-Biased Technological Change: Evidence from the U.S.*, August 2025.

⁷² Institut de l'Entreprise and McKinsey, *AI and the Evolution of Skills in France*, January 2025.

⁷³ See in particular the overview of AI in healthcare in France from February 2025, Microsoft Word - 20250205_Etat_des_lieux_IA en santé_V7.docx.

applications, which can now easily do their homework for them, including complex tasks requiring creativity.

By focusing almost exclusively on higher education in AI, the SNIA has so far failed to enable France to prepare for the scale of the upheavals that AI will bring to all professions and training programmes. The “Skills and Jobs of the Future” programme has not been sufficiently framed by an analysis of skills needs and the changes that need to be made to the existing training offer. Awareness of the changing skills needs in the labor market brought about by the accelerating pace of the ongoing technological revolution remains low today.

Nothing structured or commensurate with the challenges has yet been devised or implemented to address the “training wall” that these upheavals imply: transformation of initial training (school and higher education) in all subjects and fields, each of which is impacted differently by AI, both in terms of preparing young people for new or future labor market expectations and in terms of the impact of AI on teaching methods; an unprecedented effort in continuing education and support for professional change to adapt the workforce to the professional developments brought about by the adoption of AI in all economic sectors, including for white-collar workers.

Given the scale and speed of the technological revolution underway, several projects appear urgent and unavoidable⁷⁴ :

- map changes in skills and the job market. The development of an employment-skills outlook” in AI is included in the roadmap of the digital programmes agency led by INRIA in close collaboration with universities;
- review the content of all higher education curricula (at all levels and in all disciplines) in order to adapt each course to the new needs resulting from the spread of AI in the economy. These reviews must be conducted under the dual supervision of AI experts and specialists in each field. Such joint teams could also be set up in the regions (INRIA regional centers – universities – economic sectors) in order to take account of regional dynamics;
- define and implement an initial and continuing training programme for all teachers in the sectors concerned. Training trainers will be an essential first step, by drawing in particular on the increasingly varied and high-quality range of certified training courses available free of charge online;
- end the “Skills and Jobs of the Future” scheme as it currently exists, to give it a more strategic framework, drawing on feedback from funded projects;
- disseminate AI in all areas of research, promoting the development of AI tools, particularly sovereign ones, tailored to the needs of researchers;
- define an introductory programme on AI and its challenges in schools;
- define a programme for adapting teaching tools and methods in schools and universities to AI, which promotes intellectual development and, in particular, rethinks assessment methods and homework;
- define a policy to accelerate continuing education in companies, including through the use of incentive mechanisms, for professions whose content is most transformed by artificial

⁷⁴ See recommendation no. 6.

intelligence; discussions, initiated as part of the preparation of the European “Apply AI Strategy” of October 2025, on a personal training account dedicated to AI and digital skills, and a programme to encourage employers to accelerate continuing education on artificial intelligence, are moving in this direction⁷⁵ ;

- define a policy to anticipate and support professional change by covering part of the risk borne by workers.

Under the impetus of the new SGIA, INRIA’s digital programmes agency, in its role of steering higher education and research, is called upon to play a key role in the implementation of these projects, in close partnership with the relevant ministries (Ministry of the Economy, Finance and Industrial and Energy Sovereignty, Ministry of Higher Education and Research, Ministry of Labor, Health, Solidarity and Family) and in conjunction with the CSF “Trusted Software and Digital Solutions”. Consultation between the government and the main players[<] in continuing vocational training, namely the professional branches, skills operators and regions, is also necessary.

B - Accelerating the adoption of AI by businesses

The ability of companies to adopt reliable AI solutions tailored to their needs has become a key factor in their competitiveness in the market. At the macroeconomic level, the speed and quality of AI adoption by companies can contribute significantly to growth – much more so than the growth of AI companies alone, which has been the SNIA’s economic priority to date. A survey conducted in June 2025 by Bpifrance among SME and mid-cap company executives shows that 58% consider AI to be a matter of survival for their company, 43% have adopted an AI strategy, but only 32% are already using AI⁷⁶ .

AI systems offer increasingly varied and powerful use cases, which can: on the one hand, with generic functionalities, automate and increase the performance (efficiency and effectiveness) of cross-functional tasks or support functions such as accounting and finance, human resources management, purchasing, marketing, customer management, sales, and after-sales service; on the other hand, with more specific developments, profoundly transform the very core of goods and services production processes and, upstream, essential functions such as research and development, investment, new product development, etc.

To be successful, this adoption must be carried out in a highly professional manner⁷⁷: it requires in-depth knowledge of AI offerings and their rapid evolution, an analysis of the company’s internal needs and capabilities, an assessment of the benefits of available AI solutions but also the risks they present, compliance issues and the adjustments to be made to ensure the effective use of the chosen solution, control over its deployment, high-quality communication in the context of social dialogue, and support for those whose work is impacted.

⁷⁵ See, for example, proposals 1 and 2 of the [Hacktivate AI report](#), which emerged from a hackathon on public policy on AI organized by Allied For Startups and OpenAI, in which Mefsin participated.

⁷⁶ Bpifrance, *French companies and AI: the dawn of a revolution*, June 2025. The study estimates that the successful spread of AI throughout the economy would result in at least 1.3 additional GDP points per year by 2034.

⁷⁷ A recent study of the Massachusetts Institute of Technology (MIT) highlighted the cost and risks associated with companies purchasing AI systems that are insufficiently adapted to their operational context and adaptable based on feedback, cf. [MIT Nanda, The State of AI in Business 2025, The GenAI Divide, July 2025](#).

Specific consulting services appear to be essential, bringing together AI specialists and professionals from the client company, which SMEs and mid-cap companies do not always have the time or resources to seek out.

To date, the SNIA has not made the adoption of AI by businesses a central focus (the AI Booster scheme has remained small in scale and limited in impact), even though the rapid spread of AI in the economy appears to be crucial for the coming years, both at the microeconomic level in the race for performance (or even survival) of companies operating in highly competitive markets, and at the macroeconomic level for potential productivity gains, growth, exports and employment, even if some effects will only fully materialize in the medium term, as AI is integrated into value chains.

Several constraints come into play: funding from France 2030, which focuses on supply-side issues, does not facilitate a demand-side orientation, which also presents a high risk of windfall effects; when use cases have not already been resolved, the integration of AI into sectoral technologies constitutes a systemic rather than incremental disruption, requiring rapid-response experimental laboratories (cycles of three to six months, rather than 18 months as in traditional calls for projects); the goal of adoption by companies, which are looking for the most efficient and least risky solution on the AI market, may conflict with that of sovereignty, particularly with regard to cloud-based solutions⁷⁸. Finally, the issue of compliance with current regulations, including the new AI Act, may also be an obstacle.

The next five years will be critical. It seems essential to massively increase the adoption of AI by businesses. Under the authority of the new SGIA, several initiatives, some of which are included in the announcements made around the third phase of the SNIA, could contribute to this⁷⁹ :

- communicating with companies in all sectors to raise awareness of the challenges of adopting AI and the benefits of using sovereign solutions to limit the risks of technological and commercial dependence. All business, regional and sectoral networks should be mobilized. For its part, Bpifrance could launch a door-to-door campaign similar to the one it deployed for the “green industry” action plan;
- promote and make available to businesses resolved use cases, whether open source or not; cases related to support functions should be particularly highlighted to businesses, as they are fruits within reach;
- support pioneering companies with unresolved use cases, particularly those affecting core business processes, and, if successful, publish the results online;
- create a voluntary programme in French companies for the adoption of AI, open to young graduates with specialized training in AI, based on the model of “international voluntary work in companies”;

⁷⁸ In its aforementioned report, the AI Commission emphasizes that the adoption of generative AI is largely linked to the use of clouds by companies, with major cloud providers offering AI services and facilitating experimentation. However, France is one of the European countries where businesses use the cloud the least (30% of businesses in France in 2024, compared to more than 70% in the Nordic countries, 60% in Ireland and 45% in Germany).

⁷⁹ See recommendation no. 7.

- continue to nurture and promote the free online training and advice on issues related to the adoption of AI by businesses and, in partnership with the CNIL, on compliance with current data protection regulations;
- more generally, communicate on the private offer of personalized support for SMEs and mid-cap companies in the adoption of high-quality AI solutions that comply with regulations;
- promote, as a lever for in-depth transformation, the integration of skills and even the acquisition of AI startups by large French companies and mid-cap companies;
- promote the development of high-performance software associated with sovereign clouds, in particular by encouraging partnerships with large French companies using these technologies;
- confirm the implementation of a system of subsidized loans for companies investing in AI solutions, while ensuring that the risk of windfall effects is limited;
- accelerate the implementation of INESIA to strengthen confidence in the economic environment.

C - Invest in data, its access, quality, protection and its sovereign storage

AI systems require large amounts of data for both training and deployment. The conditions for accessing this data, particularly personal data, its quality and relevance, the security of its use, and the conditions for its storage raise complex conceptual, practical and legal questions. In its aforementioned report, the AI Commission analyzed these issues in detail and called for “*transforming our approach to personal data to better innovate*”.

Upstream of the AI value chain, the development of data storage capacities, including cloud storage, also presents several challenges. 86% of global data center capacity is currently located in the United States (54 GW of installed capacity) or China (32 GW), far ahead of the EU (12 GW)⁸⁰. As noted by the Competition Authority, the cloud sector in France is currently dominated by three American companies, which combine their storage services with generative AI tools, a situation that raises questions of sovereignty and competition⁸¹. As part of the France 2030 programme, a specific acceleration strategy is dedicated to cloud development. At EU level, research, development and industrial deployment of advanced cloud technologies are also the subject of a support programme. Private investment in data centers is growing, particularly in Marseille, town which is strategically located due to the convergence of submarine cables to America and Asia.

In the age of AI, data is the “new gold”. This requires a data management policy, the only component of the AI value chain that remains in the long term, beyond the evolution of

⁸⁰ Microsoft AI Economy Institute, *AI Diffusion Report: where AI is most used, developed and built*, November 2025.

⁸¹ Competition Authority, Opinion 23-A-08 of 29 June 2023.

algorithms and computing resources. Under the leadership of the new SGIA, France's public policy on AI must address this issue in its entirety⁸²:

- accelerate the opening up of access to data for the training and deployment of AI systems. New avenues could be explored, in particular by favoring sectoral and federated approaches. In the field of health in particular, defining an operational governance framework requires a new form of R&D to avoid the risk of developing a centralized approach that is detached from the organization of healthcare provision, which could quickly become obsolete and raise issues of sovereignty;
- ensure the quality of this data in terms of reliability, structure, relevance and suitability for purpose. This means promoting links between AI developers and data holders, including linguistic, cultural and heritage data;
- modernize the doctrine of protected data management (personal, intellectual property, copyright, cultural, heritage, etc.), taking better account of the upheavals brought about by AI and creating a better balance between user rights and business needs. This requires exploring new avenues, in particular by: better differentiating according to the nature and sensitivity of the data concerned; encouraging the use of Privacy Enhancing Technologies (PETs); developing collective and collaborative approaches to data protection; studying the advisability, in priority sectors such as health, of adopting ex post verified compliance declaration procedures for training systems, and making a priori openness the norm, encouraging the use of the legal basis of legitimate interest in the implementation of AI systems;
- accelerate investment in storage capacity in the country, taking advantage of France's competitive advantages (including nuclear energy, submarine cable network) and promoting private investment and new forms of public-private partnerships, while ensuring that the operating conditions of these centers are favorable to the French ecosystem;
- develop governance of data of public interest which, far from pitting regulation against innovation, continues to build a mutually fruitful dialogue between research, innovation and regulatory authorities, in particular the CNIL. More broadly, strategic thinking on data must involve all stakeholders under the authority of the new SGIA and integrate the various dimensions of the issue.

D - Build a realistic ambition in terms of components and better integrate “enabling” issues

A fourth critical challenge concerns the need to build realistic ambitions upstream of the AI value chain (electronic components such as accelerators and specialized chips), an issue to be monitored in connection with France's 2030 electronics acceleration strategy, and more generally to better integrate all the “enabling” issues of AI, data, computing capabilities, but also fiber optic networks, energy supply and infrastructure environment, regulation, investment capacity in AI companies, etc.⁸³

⁸² See recommendation no. 8.

⁸³ See recommendation no. 9.

On the first of these fronts, the challenge is to promote the emergence of a French (and European) electronics industry adapted to AI, an area in which France lags significantly behind. However, everyone agrees that it would be unrealistic to try to catch up with the giants in the production of graphics processors (GPU chips). In its aforementioned report, the AI Commission therefore considers that “*it is not a question of chasing technological advances, but of creating our own comparative advantages*” and thus targeting certain segments of the value chain.

Expanding public policy on AI under the authority of the new SGIA to include a realistic and clear ambition for the upstream part of the AI value chain (hardware and software components for high-performance computing in AI) would involve, in particular:

- building a synchronized industrial strategy for algorithms, software platforms and components for AI (including those required for computing infrastructures), with a comprehensive vision of the building blocks needed for strategic autonomy in various industrial sectors;
- linking this policy to a strategy for developing a European AI components sector. In particular, data and computing centers that receive EU co-financing should be encouraged to increase the share of French and European hardware components;
- in the short term, in order to catch up, develop targeted partnerships with major tech companies and international sovereign wealth funds and, beyond that, develop a strategy for international scientific collaboration and focus public support for this strategy realistically on segments of the value chain where comparative advantages can be exploited;
- in the longer term, promote research and innovation into possible disruptive technologies for the next generation of AI components;
- coordinate with competition policy, which, in a highly concentrated field, is set to play a decisive role in facilitating the emergence of new players.

On the second front, “enabling” issues, i.e. those concerning the infrastructure on which the development of AI depends, have so far been integrated into public AI policy to varying degrees, more clearly for some (e.g. computing infrastructure) and less so for others. With the establishment of the SGIA, public policy on AI would benefit from better integrating the diverse range of critical “enabling” issues. This involves, in particular:

- defining the critical needs for the development of AI in terms of electrical infrastructure and connectivity (particularly submarine cables);
- and ensuring that these critical needs are taken into account in the relevant strategies.

E - Making AI a lever for quality, efficiency and effectiveness of public action

A fifth and final critical challenge concerns the transformation of public action through AI. So far, results have been very modest in this area, which was not a priority in the second phase of the SNIA. There is a risk that the government, in its pace of AI adoption, will find itself increasingly out of step with users of public services, beneficiaries of public policies, its

suppliers, etc. For example, to what extent would it be possible for a court to continue to provide the public service of Justice without resorting to AI in the face of law firms that make intensive use of it? A recent study estimates that a large third of public sector jobs worldwide (38% in the case of France) will be substantially impacted by generative AI, which will take over peripheral tasks, introduce new functionalities or replace certain functions by automating them (in one in five cases)⁸⁴.

A “ministerial roadmap” exercise was launched in February 2025, which aims to update the digital transformation trajectories of ministerial departments by integrating AI as a priority area (evaluating experiments conducted to date and identifying those that could be scaled up). The results of this campaign, which were due to be presented in June 2025, were not presented due to a lack of satisfactory substance. The DINUM estimates that the cost of deploying a trusted infrastructure and initial generic generative AI services for all public servants, as well as scaling up the most mature AI solutions in ministries, would amount to about a hundred million euros per year from 2026 onwards, a budget that is not currently planned.

A new type of investment appears essential to make artificial intelligence a lever for quality, efficiency and effectiveness in public administrations and policies⁸⁵. Among the conditions for the successful adoption of an AI solution by a public service are the prior structuring of data, attention to the quality of digital infrastructure, the adaptation of the AI solution to specific operational needs (which may, depending on the case, involve existing solutions with minimal adjustment or require a more tailor-made solution), a proper understanding of data protection and sovereignty issues, and training for all staff directly or indirectly involved in the AI solution being introduced.

Under the authority of the new SGIA, public policy in this area would benefit in particular from:

- establishing a mechanism to accelerate the adoption of AI by public administrations (public sector AI Booster): in each ministry (or initially within ministries identified as priorities), in addition to appointing ministerial AI coordinators, a project team could be set up, bringing together AI solution specialists, ministry departments and inspection bodies to identify needs and the AI solutions that meet them; when identifying needs, priority should be given to projects with the greatest impact in terms of quality, including for users with disabilities or who encounter particular difficulties, efficiency and effectiveness of public action; when identifying AI solutions, the use of resolved use cases, particularly open source ones, would enable deployment without delay and at minimal cost; a network of ministerial AI coordinators under the responsibility of the Secretary General for AI and his team must also be established to ensure the consistency of this work and promote pooling of resources; support for the deployment of AI systems must also be put in place, including training for the civil servants concerned, as well as transparent information that feeds into social dialogue;
- further mobilizing public procurement and the expertise of French AI startups in the search for solutions adapted to public services, by making the necessary adjustments to public procurement law, particularly when a fundamental interest is at stake;

⁸⁴ Roland Berger, *The Public Sector in the Age of Gen AI*, September 2025.

⁸⁵ See recommendation no. 10, to be read in conjunction with recommendation no. 1.

- considering the introduction of a fiscal incentive mechanism for ministries to finance AI transformation projects. This mechanism could take the form of ring-fencing funds dedicated to AI transformation or even returning a portion of the savings generated by a project, as was done to boost the government's real estate policy;
- pursue the objective of strengthening the fight against tax and social security fraud through AI, while ensuring complete reliability of results and strict compliance with personal data rights and protection. The additional revenue, savings and avoided expenditure that this investment will generate should be accurately quantified and seen as a contribution by AI to the recovery of public finances;
- more generally, speed up efforts to train public officials in AI and develop, to support the momentum of AI innovations, a culture of interdisciplinary and dynamic project management that promotes interaction between all stakeholders.

CONCLUSION AND RECOMMENDATIONS

A third phase of SNIA was launched in February 2025, with the aim of accelerating the diffusion of artificial intelligence in businesses.

Artificial intelligence is no longer the same issue it was in 2018, when the SNIA was launched. The revolution brought about by this technology has reached a magnitude that few other technological breakthroughs in history can match. AI is no longer a matter for specialists alone; it affects the entire economy and society as a whole, and is becoming an essential priority for public policy.

Successfully scaling up the artificial intelligence revolution requires several prerequisites to be met: strengthening interministerial coordination of public policy on AI, in particular by setting up a dedicated AI secretariat attached to the Prime Minister's office; better integrating the challenges of public finance sustainability and AI policy effectiveness; conducting an in-depth assessment of the results of the previous phases of the national strategy and drawing inspiration from best practices abroad; improving coordination with the European level; better mobilizing the regions; and redefining the division of roles with the private sector.

It would be a mistake to consider that the priorities on which the SNIA has achieved initial success no longer require attention, given the rapid pace of change in the AI landscape. Public policy on AI must seek to expand its areas of excellence and aim for even more structural transformations and impacts. Five key areas, which have been pursued since 2018, should therefore be further developed, with the aim of anchoring the ecosystem of excellence in AI training, research and innovation, change the paradigm to strengthen computing capabilities, amplify transfers to industry and strengthen the AI sector, put artificial intelligence at the service of the common good and strengthen trust and security, and better respond to challenges related to energy needs, frugality and the sustainability of AI.

Finally, the first two phases of the SNIA have blind spots in several key areas, or at least have failed to achieve results commensurate with the challenges. This is particularly the case in projects that require the involvement of a wide range of stakeholders and links with other public policies. Five critical challenges, which have not been sufficiently addressed to date, must therefore be placed at the heart of public policy on AI in the coming years: adapting training in all sectors and anticipating changes in the labor market; accelerating the adoption of AI by businesses; investing in data, its access, quality, protection and sovereign storage; building a realistic ambition in terms of components and better integrating "enabling" issues; and making AI a driver for quality, efficiency and effectiveness in public action.

The Court makes the following recommendations:

- 1. strengthen interministerial oversight of public AI policy by the end of 2025 through the creation of an ad hoc general secretariat, and better reconcile the necessary ambition of this policy with the challenges of efficiency and effectiveness by conducting an in-depth evaluation of results and seeking greater complementarity with the European level. regions and the private sector (Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, Ministry of Higher Education and Research, INRIA);*

adopt, by the next AI summit in February 2026, a strategy aimed at:

- 2. enshrine long-term excellence in training, research and innovation in the field of AI by better defining needs, giving stakeholders greater responsibility, reaffirming the purpose of public research and promoting mobility with the private sector (Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, Ministry of Higher Education and Research, INRIA);*
- 3. increase, in coordination with the EU strategy, computing capacities for AI and redefine, through new forms of partnership between the public and private sectors, the conditions for financing, building and operating these infrastructures, which guarantee open access to all users (Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, Ministry of Higher Education and Research, INRIA, CNRS);*
- 4. strengthen the transfer of research to industrial development and support the growth of AI companies, including by leveraging public procurement and strengthening the monitoring of industrial acquisitions in the AI sector (Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, Ministry of Higher Education and Research, INRIA);*
- 5. implement and pursue commitments to trustworthy, frugal and sustainable AI, including at European and international level (Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, Ministry of Higher Education and Research, Ministry of Europe and External Affairs, INRIA);*
- 6. better anticipate changes in the job market, adapt teaching methods and tools as well as the content of all higher education courses to AI, disseminate AI in all research sectors, refocus continuing education efforts, and support professional changes related to AI (Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, Ministry of Higher Education and Research, Ministry of Labor, Health, Solidarity and Family, INRIA);*
- 7. support the acceleration and mass adoption by businesses over the next five years of AI use cases tailored to their needs, including by increasing communication, making available resolved use cases, supporting pioneering businesses on as yet unresolved use cases, and promoting the development of high-performance software associated with sovereign clouds (Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, Bpifrance, INRIA);*
- 8. in coordination with all stakeholders, strengthening access to data for AI systems, guaranteeing its quality, better protecting sensitive data and intellectual property, taking into account the latest scientific advances (federated learning, decentralized AI) and investing in sovereign storage capacities (Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, CNIL);*
- 9. build realistic ambitions for electronic components for AI and better coordinate public policy on AI with policies on the infrastructure that underpins its development, particularly with regard to electricity supply and connectivity (Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, Bpifrance, INRIA);*

10. accelerate the transformation of public administrations and policies through AI, in particular through increased use of innovative public procurement and the introduction of incentive mechanisms, support measures and specific training for civil servants (Prime Minister, Ministry of the Economy, Finance and Industrial and Energy Sovereignty, INRIA).
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General conclusion

The national strategy for artificial intelligence has created real momentum since its launch in 2018. France has managed to climb to the top of the European pack, including in the latest and most widely used generative AI technologies. It can compete with its rivals, the United States and China set apart, in most AI-related innovations, and its visibility on the international stage, as confirmed by the Paris summit in February 2025, is real. On the other hand, support for the dissemination of AI beyond the circle of specialists – businesses, public administrations, students, citizens – has so far received too little attention.

The Court of Auditors has made ten recommendations, the implementation of which would enable public policy on AI to change scale, build on the initial successes of the SNIA, and overcome the limitations and shortcomings identified.

It is on this condition that France, in close cooperation with the European Union and local authorities, drawing on the training-research-innovation ecosystem as well as businesses and investors, even if it means changing its operating methods, will continue on its path to excellence and succeed in embracing all the dimensions that the AI revolution is set to affect, in the service of the common good and with a view to guaranteeing national sovereignty.

List of abbreviations

AFNOR	French Standardization Association
AI	Artificial intelligence
AI Act.....	European Regulation on Artificial Intelligence of June 2024
AISSAI.....	AI for Science and Science for AI
ALT-EDIC	Alliance for Language Technologies
ANR	National Research Agency
ANSSI	National Cybersecurity Agency
CEA.....	French Atomic Energy and Alternative Energies Commission
CEI	Call for expressions of interest
CIAN.....	Council for Artificial Intelligence and Digital Technology
CNAM.....	National Health Insurance Fund
CNIA.....	National Coordinator/Coordination for Artificial Intelligence
CNIL	National Commission for Information Technology and Civil Liberties
CNRS	National Centre for Scientific Research
CSF.....	Strategic Sector Committee
DGE	Directorate-General for Enterprise
DIAT	Demonstrators of frugal AI serving the ecological transition in local areas
DINUM	Interministerial Directorate for Digital Technology
DITP.....	Interministerial Directorate for Public Transformation
EDPB	European Data Protection Board
ETI	Medium-sized enterprise
EU	European Union
GDPR.....	General Data Protection Regulation of April 2016
GENCI.....	Large national high-performance computing facility
IEA	International Energy Agency
INESIA.....	National Institute for the Evaluation and Security of Artificial Intelligence
INRIA.....	National Institute for Research in Computer Science and Control
IPCEI.....	Important Project of Common European Interest
LNE.....	National Metrology and Testing Laboratory
MOOC.....	Massive Open Online Course
OECD.....	Organization for Economic Co-operation and Development
PEPR	Priority Research Programme and Equipment
PET.....	Privacy Enhancing Technology
PEReN.....	Digital Regulation Expertise Centre
PIA	Future Investment Programmes
SME	Small or medium-sized enterprise

GPAI	Global Partnership on Artificial Intelligence
PNRIA.....	National Artificial Intelligence Research Programme
PRISSMA.....	Research and Investment Platform for Safety and Security in autonomous mobility
R&D	Research and development
SGDSN.....	General Secretariat for Defense and National Security
SGIA	Secretary General for Artificial Intelligence
SGPI.....	General Secretariat for Investment
SNIA	National Strategy for Artificial Intelligence
SNDS	National Health Data System
TEF.....	AI Testing and Experimentation Facility

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Appendix n° 1 : Brief glossary

Artificial intelligence (AI): a machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments⁸⁶.

Types of AI systems

Machine learning: a set of methods that enable a system to ‘learn’ to perform tasks from data, i.e. to improve its performance in solving tasks without being explicitly programmed for each one.

The development of an AI system involves several phases: planning and design; data collection and pre-processing, involving *Text and Data Mining (TDM)*, including harvesting methods (scraping), i.e. the automated retrieval and organization of data available on the internet; training or *learning*, a phase that consists of determining the model by having it solve practical tasks based on the data collected (called observations); testing, evaluation, verification, validation, and *fine-tuning*, in particular with a view to specializing the pre-trained model to perform a specific task; production or *inference*, a phase in which new data can be submitted to the model in order to obtain the result corresponding to the desired task, while minimizing statistical errors; this phase also involves deploying the system to users and managing it over time (operation, maintenance and monitoring); withdrawal and dismantling (and often replacement with a more efficient version).

We refer to *unsupervised learning* when the algorithm seeks to discover underlying structures in the data, and *reinforcement learning* when the model learns incrementally, based on a "reward" received for each successful action.

Artificial Neural Networks (ANN): a model inspired by the structure and function of biological neural networks in animal brains. Each unit (artificial neuron) receives signals from connected neurons, processes them and sends a signal to other connected neurons. The signal is calculated by an activation function that depends on the signals received and whose intensity is determined by a weight that adjusts during the learning process. Artificial neural networks are one of the techniques used for machine learning.

Deep learning: a field of AI that uses artificial neural networks forming numerous superimposed layers to solve complex tasks. Deep learning enables machines to learn automatically from large amounts of data, extracting complex features to solve classification, prediction and recognition problems. Deep learning enables significant advances in sound and visual signal analysis (facial and voice recognition, automated language processing, etc.). Deep learning is one of the applications of artificial neural networks.

Large Language Model (LLM): a deep learning AI system designed to process vast amounts of data (typically in the order of a billion or more) in order to understand and generate text.

⁸⁶ See Article 3 of the European Regulation on Artificial Intelligence of June 2024.

Small Language Model (SLM): in contrast to LLMs, SLMs are AI systems that use deep learning techniques that require less data and computing power to operate, making them more practical for less energy-intensive use in devices with limited resources or for applications requiring rapid response.

Chatbot: software designed to interact with users through text or voice exchanges. LLMs and SLMs have enabled rapid progress in this area, to the point where AI is sometimes confused, in the minds of the general public, with these chatbots alone.

Generative AI (GenAI): An AI system capable of generating text, images, videos or other media in response to queries, using underlying data structures to produce new data. This type of system relies on deep learning techniques. LLMs and SMLs are examples of generative AI.

General Purpose AI (GPAI): an AI model that exhibits significant generality and is capable of competently performing a wide range of distinct tasks and can be integrated into a variety of downstream systems or applications⁸⁷, as opposed to narrow AI, which is limited to specific tasks. This concept should be distinguished from that of *Artificial General Intelligence (AGI)*, a hypothetical type of AI that would be highly autonomous and equal or surpass human capabilities in most or all economically valuable cognitive tasks.

The model is said to be “open” when its parameters, including weights, information about the model architecture and information about the use of the model, are made public⁸⁸.

The term “*foundation models*” is also used to refer to a machine learning model that has been pre-trained on a particularly large and diverse dataset, has general capabilities, and can be adapted to a wide variety of tasks.

Characteristics of AI systems

Trustworthy AI: an AI system in which all stakeholders have confidence throughout its life cycle. The June 2024 AI Act sets out three principles: legality of the system (compliance with applicable laws and regulations), ethics (respect for principles such as human autonomy, prevention of harm, fairness, explainability) and robustness (technical reliability, minimizing the risk of errors or abuse).

Frugal or sober AI: an approach to AI that seeks to minimize the environmental footprint of models while maintaining high performance. One of the techniques used is *knowledge distillation*, which involves transferring knowledge from a large model (teacher) to a smaller (student) model that is architecturally simpler, by training it to imitate the initial model.

Embedded AI: integration of AI algorithms into autonomous devices with limited resources, such as connected objects, vehicles or robots.

Decentralized AI: an AI model that does not rely on a single entity but uses distributed networks to process information and algorithms that can run on multiple nodes in a network, allowing for greater flexibility and resilience in the face of failures or cyberattacks.

⁸⁷ See Article 3 of the European Regulation on Artificial Intelligence of June 2024.

⁸⁸ See recital 102 of the European Regulation on Artificial Intelligence of June 2024. Defining what constitutes an open model remains a challenge, however.

AI applications: the use of AI is spreading, in various forms, throughout the economy and academia. Until now, the diffusion of AI has mainly taken the form of weak AI systems, i.e. programmes developed to perform specific tasks.

General-Purpose Technology (GPT): Like printing, electricity and computers, AI is considered a general-purpose technology, with implications for a very wide range of activities, transforming them significantly and, more broadly, potentially affecting the entire of society.

Tools used by AI systems

Graphics Processing Unit (GPU): a processor optimized for parallel data processing. Initially designed for graphics rendering, GPUs have become central to AI, particularly for deep learning. They are an essential part of the physical components (*hardware*) required for AI, with a view to developing *AI software* and computer applications. *Middleware*, third-party software that connects different applications, also plays a key role in optimizing complex AI systems.

Data center: a collection of equipment (mainframe computers, servers, storage arrays, networks and telecommunications, air conditioning, *etc.*) for the storage, security and management of computer data. *Sandboxing* is a security mechanism implemented by an operating system to isolate data development or analysis, remaining disconnected from the decision-making information system so as not to risk compromising the main information system.

Cloud computing (Cloud): the practice of using computer servers hosted in data centers connected to the Internet to store, manage and process data, rather than a local server or personal computer. There are three types of *cloud*: the *public cloud*, accessible via the Internet; the *enterprise or private cloud*, accessible only on a private network; and the *intermediate or hybrid cloud*, which is a combination of the two. The main services offered in the cloud are: *SaaS (Software as a Service)*, where the customer uses software installed on remote servers rather than on the user's machine; *PaaS (Platform as a Service)*, mainly intended for developers, where the customer retains control of the applications themselves and the cloud provider offers a platform for running these applications, which may include hardware, basic software and the infrastructure for network connection, storage, backup, *etc.* ; and *IaaS (Infrastructure as a Service)*, where the customer manages the application software and *middleware*, and the cloud provider manages the server hardware, virtualization layers, storage and networks.

High-Performance Computing (HPC): a computer designed to achieve the highest possible performance, particularly in terms of AI computing capabilities. There are three components: hardware (electronic design of the tool), software (software adaptation to the tool) and user support (network of engineers and technicians to operate and optimize the tool). The number of floating-point operations per second (*FLloating-point Operations Per Second*, flop/s) is a unit of measurement for the computing speed of a computer system. *Exascale computing* refers to supercomputers with a computing power exceeding 10^{18} flop/s (and *petascale computing* for 10^{15} flop/s). Cloud computing can be associated with these high-performance computing capabilities (known as *Cloud HPC*).

AI and disruptive innovation (*Deeptech*)

Players in disruptive technological innovation or *deep tech* are distinguished by new solutions to complex problems, pushing the boundaries of knowledge and innovation, transforming industries and creating new markets. Three main criteria distinguish a company, often a start-up, from *deep tech*: it develops in close connection with research (generally over long cycles, and with corporate governance that remains closely linked to the scientific or technological world), which requires costly investments before achieving commercial profitability (with a major role played by venture capitalists able to invest in projects with a strong technological component); it exploits disruptive technology, which is often protected and difficult to reproduce (intellectual property, industrial property and specific know-how, which can create barriers to entry for competitors); it leads to industrial and even societal changes, generally giving rise to a new category of product or service with high added value and differentiation, which may ultimately replace a dominant technology in a market or create a new market.

For a more detailed French glossary on AI, readers may find it useful to refer to the one developed by the French Data Protection Authority (CNIL)⁸⁹.

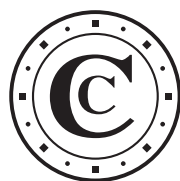
⁸⁹ [Glossary of artificial intelligence \(AI\) | CNIL](#).

Appendix n° 2 : Follow-up to the recommendations of the Court of Auditors' April 2023 report on the research component of the first phase of the SNIA

<i>Recommendations made</i>	Assessment by the Court of the degree of implementation	Comments
Translate public policy on artificial intelligence into a summary fiscal document that will enable it to be monitored and its effects measured its effects (<i>Ministry of the Economy, Finance and Industrial and Energy Sovereignty</i>).	Partial implementation	A fiscal monitoring document for the SNIA exists within the framework of France 2030 and is shared between ministries. It is regularly consolidated by the coordinator in conjunction with the SGPI. However, there is no document listing all public efforts in favor of AI.
Specify the respective missions of the 3IA and non-3IA centres of excellence, and clarify the multi-year funding allocated to them accordingly (<i>Ministry of Higher Education and Research, SGPI</i>)	Implementation	The certification of AI clusters provided an opportunity to clarify the identification and missions of centers of excellence. Multi-year funding has also been clarified and extended to five years.
Establish shared objectives and priority indicators for public policy on AI, in line with the European strategy (Ministry of the Economy, Finance and Industrial and Energy Sovereignty, SGPI).	Implementation	As part of France 2030, four indicators have been defined to measure the structural effects of the SNIA on AI training, innovation, attractiveness, and the dissemination of AI.
Create a scientific and steering committee at INRIA, co-chaired by France Universités, to monitor the implementation of the strategy in a concerted manner and define future strategic orientations (<i>Ministry of Higher Education and Research, INRIA</i>)	Implementation	The establishment of a strategic committee for the Higher Education and Research programme, meeting monthly, is one of the explicit tasks of INRIA's new mission statement for the national coordination of the Higher Education and Research component of the strategy. This committee was set up in autumn 2024.
Produce a harmonized and up-to-date map of AI training courses to be promoted through a common label to make them visible and support their massification (Ministry of Higher Education and Research)	Partially implemented	There are still several maps of AI training courses that are not federated or based on uniform criteria. However, the French School of Artificial Intelligence label, promoted by the 3IA institutes as part of the CEI "Skills and Jobs of the Future", nevertheless makes it easier to identify and increase the visibility of training programmes.
Forecasting secondary school teacher requirements for secondary schools, teacher-researchers and researchers trained in the use of AI, and establish appropriate training plans (Ministry of Higher Education and Research)	Not implemented	
Develop a charter and catalogue of best practices to define and monitor the environmental impact of AI research and	Implemented	During the second phase, an environmental impact assessment was added to each SNIA measure. The

<i>Recommendations made</i>	Assessment by the Court of the degree of implementation	Comments
promote the development of responsible AI (SGPI, Ministry of Higher Education and Research)		challenge now is to improve the monitoring of this impact.

Source: Court of Auditors



France developed a national strategy for artificial intelligence in 2018.

The first phase, running until 2022, involved €1.3 billion in government funding and enabled the structuring of research and an ecosystem of innovation in artificial intelligence.

The second phase, running until 2025, in which the government has invested €1.1 billion, has consolidated the structuring of centers of excellence and enabled France to achieve initial successes in generative artificial intelligence, in the fields of frugality and trust, and to play an active role on the European and international stages. However, the challenge of mass adoption and support for the diffusion of artificial intelligence beyond the circle of specialists has not yet received sufficient attention.

A third phase of the national strategy was launched in February 2025, but several prerequisites must be met in order to achieve the scale change required by the artificial intelligence revolution.

The successes of the first two phases must be taken further to achieve a more structural transformation. Five critical challenges, which have not been adequately addressed to date, must be placed at the heart of public policy on artificial intelligence. The Court of Auditors has made ten recommendations, the implementation of which would enable the initial successes to be built upon and the limitations and shortcomings identified by the Court to be overcome.